

(...), then {...} and then [...].

* For modulus e.g. $|-3|$ we write magnitude only not sign i.e. 3 in this case.

Important Formulas

- 1). $(a + b)^2 = a^2 + 2ab + b^2$
- 2). $(a - b)^2 = a^2 - 2ab + b^2$
- 3). $(a^2 - b^2) = (a + b)(a - b)$
- 4). $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- 5). $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- 6). $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
- 7). $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
- 8). $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$
Here, if $(a + b + c) = 0$,
then $a^3 + b^3 + c^3 - 3abc = 0$
 $\Rightarrow a^3 + b^3 + c^3 = 3abc$
- 9). $(a + b + c)^2 = (a^2 + b^2 + c^2) + 2(ab + bc + ca)$

Variety Questions

Q.1. The simplified form of the expression

$$\frac{(x^2 - 9) \times (9x^2 - 1)}{(x - \frac{1}{3}) \times (1 - \frac{x}{3})} \times \frac{9}{(x + \frac{1}{3})(1 + \frac{x}{3})}$$

is _____.

SSC CHSL Tier II (10/01/2024)

- (a) -729 (b) -81 (c) 729 (d) 81

Q.2. If $A = \frac{58^2 - 25^2}{46^2 - 37^2}$, $B = \frac{26^2 - 15^2}{56^2 - 15^2}$,

then the value of $\frac{1}{B} - \frac{20}{A}$ is :

SSC CGL Tier II (26/10/2023)

- (a) 2 (b) 1 (c) -1 (d) 0

Q.3. On comparing the following two numeric expressions, we find that _____.

$$[(\frac{7}{9})^2]^{\frac{3}{5}} \text{ and } [(1\frac{2}{3})^9]^{\frac{3}{5}}$$

SSC MTS 14/09/2023 (2nd Shift)

- (a) both the expressions are equal
(b) the first expression is smaller than the second
(c) the first expression is larger than the second
(d) the given two expressions cannot be compared

Q.4. On simplifying the following expression, we get _____.

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

SSC MTS 08/09/2023 (2nd Shift)

- (a) 0 (b) 3 (c) 1 (d) 2

Q.5. Simplify :

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

= ?

SSC CHSL 04/08/2023 (1st Shift)

- (a) 6.125 (b) 8.645 (c) 7.247 (d) 10.245

Q.6. Simplify the following expression.

$$0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 +$$

$$0.044 + 0.136 + 0.44 \times 0.68$$

SSC CHSL 03/08/2023 (1st Shift)

- (a) 0.69 (b) 1 (c) 0.8 (d) 0.9

Q.7. Simplify the given expression .

$$\frac{0.09 \times 0.09 + 0.04 \times 0.04 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

SSC CHSL 02/08/2023 (4th Shift)

- (a) 0.38 (b) 0.24 (c) 0.32 (d) 0.29

Q.8. What is the value of the given Expression ?

$$\frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$$

SSC CGL 24/07/2023 (1st shift)

- (a) 16 (b) 64 (c) 20 (d) 24

Q.9. What is the value of $(2^2 + 1)(2^4 + 1)$

$$(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1) ?$$

SSC CHSL 14/03/2023 (2nd Shift)

- (a) $\frac{2^{256} - 1}{2}$ (b) $\frac{2^{256} - 1}{3}$
(c) $2^{256} - 1$ (d) $\frac{2^{256} - 1}{4}$

Q.10. Which of the following statement is true ?

I. $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$

II. $\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} > \frac{7}{13}$

SSC CHSL 13/03/2023 (4th Shift)

- (a) Only I (b) Both I and II
(c) Only II (d) Neither I nor II

Q.11. Find the value of the following expression.

$$[\frac{5}{8} - \{\frac{3}{8} - (\frac{5}{8} - \frac{3}{8})\}] \text{ of } 8.8 - 1.2$$

$$4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + (\frac{3}{4} - \frac{3}{8})$$

SSC CPO 11/11/2022 (Morning)

- (a) $5\frac{22}{43}$ (b) $3\frac{23}{67}$ (c) $4\frac{44}{85}$ (d) $4\frac{4}{5}$

Q.12. Solve the following.

$$24 \div \frac{3}{8} \text{ of } (8 + 2 \times \frac{7}{3}) + [\frac{2}{11} \div \frac{4}{55} - \{\frac{5}{8} + \frac{6}{16}\}]$$

$$32 \div 15 - 7 + 75 \div (6 + 15 \div 3 + 4)$$

SSC CPO 09/11/2022 (Morning)

- (a) $\frac{23}{27}$ (b) $\frac{9}{2}$ (c) $\frac{11}{18}$ (d) $\frac{15}{19}$

Q.13. Solve

$$\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)}$$

SSC CHSL 07/06/2022(Afternoon)

- (a) 15.1 (b) 15.3 (c) 15.2 (d) 15.4

Q.14. Simplify the following expression.

$$9\frac{1}{5} \div \{ (16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3}) +$$

$$17\frac{1}{2} \text{ of } 3\frac{1}{21} \}$$

SSC CHSL 26/05/2022(Evening)

- (a) $\frac{139}{881}$ (b) $\frac{137}{881}$ (c) $\frac{140}{881}$ (d) $\frac{138}{881}$

Q.15. The value of

$$\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)} \text{ is:}$$

SSC CGL Tier II (29/01/2022)

- (a) $\frac{3}{20}$ (b) $\frac{3}{10}$ (c) $\frac{7}{20}$ (d) $\frac{9}{20}$

Q.16 The simplified value of

$$\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - 3 - 8)\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6}$$

is:

SSC MTS 11/10/2021(Morning)

- (a) $-\frac{111}{11}$ (b) $-\frac{127}{27}$ (c) $\frac{127}{43}$ (d) $\frac{111}{11}$

Q.17. If $\frac{1}{4 - \sqrt{8}} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}}$

= $a + b\sqrt{2}$, then what is the value of $(3a + 4b)$?

SSC CGL Tier II (15/11/2020)

- (a) $99\frac{1}{2}$ (b) 97 (c) $98\frac{1}{2}$ (d) 98

Q.18. If the following interchanges are made in signs and numbers, which equation would be correct ?

Interchanges: Signs: $\div$ and $-$ Numbers: 15 and 5

SSC CHSL 13/10/2020 (Morning)

- (a) $4 \times 30 - 15 \div 5 + 12 = 21$
(b) $12 \times 30 - 15 \div 5 + 4 = 37$
(c) $4 \times 30 - 15 \div 5 + 12 = 27$
(d) $4 \times 30 - 5 \div 15 + 14 = 21$

Q.19. If $\frac{4}{1 + \sqrt{2} + \sqrt{3}} = a + b\sqrt{2} + c\sqrt{3}$

$- d\sqrt{6}$, where a, b, c, d are natural numbers, then the value of $a + b + c + d$ is:

SSC CHSL 18/03/2020 (Morning)

- (a) 0 (b) 2 (c) 4 (d) 1

Q.20.

If $[(\frac{2}{3})^3]^{(2x+3)} \div \frac{-3}{4} = [(\frac{2}{3})^{\frac{2}{3}}]^{(3x+7)} \div \frac{-6}{5}$

then the value of $\sqrt{2 - 42x}$ is:

SSC CHSL 18/03/2020(Morning)

- (a) 5 (b) 6 (c) 3 (d) 4

Q.21. If '+' means ' $\div$ ', ' $\div$ ' means '+', ' $\times$ ' means ' $\div$ ' and ' $\div$ ' means ' $\times$ ' then the

value of $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$ is:

SSC CGL 03/03/2020 (Morning)

(a) $-\frac{5}{3}$ (b) $-\frac{15}{19}$ (c) $\frac{15}{19}$ (d) $\frac{5}{3}$

Q.22. The value of

$$\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$$
 is:

SSC CGL Tier II (13/09/2019)

(a) 24.42 (b) 24.24 (c) 25.42 (d) 25.48

Q.23. The value of

$$\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$$
 is:

SSC CGL Tier II (12/09/2019)

(a) -1 (b) 1 (c) 3 (d) -3

Q.24. The value of $22.\bar{4} + 11.5\bar{6}7 - 33.5\bar{9}$ is

SSC CGL Tier II (11/09/2019)

(a) $0.\bar{3}2$ (b) $0.\bar{4}1\bar{2}$ (c) $0.\bar{3}1$ (d) $0.\bar{4}1\bar{2}$

Q.25. Find the value

$$\left\{ \frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)} \right\}$$

SSC MTS 07/08/2019 (Morning)

(a) $\frac{25}{2}$ (b) $\frac{19}{2}$ (c) $\frac{23}{2}$ (d) $\frac{21}{2}$

Q.26. If $x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15}$

$$..... + \frac{1}{23.24}, y = \frac{1}{36.37} + \frac{1}{37.38} +$$

$$\frac{1}{38.39} + \frac{1}{71.72}$$
 then $\frac{x}{y}$ is equal to:

SSC CHSL 10/07/2019 (Evening)

(a) $\frac{1}{3}$ (b) $\frac{1}{24}$ (c) $\frac{1}{72}$ (d) 3

Q.27. The simplified value of

$$\frac{0.01404}{24^2 + 6^2 - 144}$$
 is:

SSC CHSL 10/07/2019 (Evening)

(a) 3×10^{-5} (b) 6×10^{-5}
(c) 2.4×10^{-4} (d) 3×10^{-4}

Q.28. The simplified value of

$$\frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95}$$
 is:

SSC CHSL 09/06/2019 (Evening)

(a) 1.0025 (b) 1.0525
(c) 1.0005 (d) 1.0505

Q.29. The value of $\frac{9}{15}$ of $(\frac{2}{3} \div \frac{2}{3})$ of $(\frac{3}{2}) \div (\frac{3}{4} \times \frac{3}{4} \div \frac{3}{4})$ of $(\frac{4}{3})$ of $(\frac{5}{4} \div \frac{5}{2} \times \frac{2}{5})$ of $(\frac{4}{5})$ is:

SSC CGL 06/06/2019 (Afternoon)

(a) $\frac{20}{9}$ (b) $\frac{4}{25}$ (c) $\frac{18}{125}$ (d) $\frac{40}{9}$

Q.30. The value of $(5 + 3 \div 5 \times 5) / (3 \div 3 \text{ of } 6)$ of $(4 \times 4 \div 4 \text{ of } 4 + 4 \div 4 \times 4)$ is

SSC CGL 06/06/2019 (Morning)

(a) $8\frac{1}{5}$ (b) $7\frac{1}{3}$ (c) $9\frac{3}{5}$ (d) $6\frac{2}{3}$

Q.31. The value of $2\frac{7}{8} \div (3\frac{5}{6} \div \frac{2}{7})$ of $2\frac{1}{3} \times [(2\frac{6}{7} \text{ of } 4\frac{1}{5} \div \frac{2}{3}) \times \frac{5}{9}]$ is:

SSC CGL 04/06/2019 (Evening)

(a) $\frac{1}{4}$ (b) 4 (c) $\frac{1}{23}$ (d) 5

Q.32. The value of $2 \times 3 \div 2$ of $3 \times 2 \div (4 + 4 \times 4 \div 4 \text{ of } 4 - 4 \div 4 \times 4)$ is:

SSC CGL 04/06/2019 (Morning)

(a) 8 (b) 1 (c) 4 (d) 2

Q.33. $\frac{5.75 \times 5.75 \times 5.75 + 3.25 \times 3.25 \times 3.25}{57.5 \times 57.5 + 32.5 \times 32.5 - 57.5 \times 32.5}$ is equal to:

SSC CPO 12/03/2019 (Evening)

(a) 0.009 (b) 0.0009 (c) 0.9 (d) 0.09

Practice Questions

SSC CHSL 2023 Tier - 2

Q.34. If $x = (\frac{3}{2})^2 \times (\frac{2}{3})^{-4}$, find the value of x^{-2} .

SSC CHSL Tier II (02/11/2023)

(a) $(\frac{3}{2})^6$ (b) $(\frac{3}{2})^{12}$
(c) $(\frac{2}{3})^{12}$ (d) $(\frac{2}{3})^6$

Q.35. If

$$\sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q}$$

then find the value of $\frac{p}{q}$.

SSC CHSL Tier II (02/11/2023)

(a) $8^3 \times 10^{-8}$ (b) $8^3 \times 10^{-9}$
(c) $8^4 \times 10^{-9}$ (d) $8^4 \times 10^{-8}$

Q.36. Simplify:

$$\left(\sqrt[3]{\sqrt[3]{343}}\right)^3 \times \left(\sqrt[3]{\sqrt[3]{216}}\right)^3$$

SSC CHSL Tier II (10/01/2024)

(a) 56 (b) 42 (c) $\sqrt[3]{42}$ (d) 42^3

Q.37. Find the value of

$$\sqrt{625} + \sqrt{\frac{625}{100}} + \sqrt{\frac{625}{10000}} + \sqrt{\frac{625}{1000000}}$$

SSC CHSL Tier II (10/01/2024)

(a) 27.075 (b) 27.775
(c) 27.75 (d) 27.025

SSC CGL 2023 Tier - 2

Q.38. What is the value of $[\frac{1}{8} + \{\frac{1}{6} \times (\frac{36}{45} \div \frac{24}{25}) - (\frac{12}{21} \times \frac{14}{15} \div \frac{24}{45})\} + \frac{27}{36}]$?

SSC CGL Tier II (26/10/2023)

(a) $\frac{1}{36}$ (b) $\frac{1}{72}$ (c) $\frac{1}{108}$ (d) $\frac{1}{27}$

Q.39. Find the value of $\frac{\sqrt{1.24} \times \sqrt{2.79}}{\sqrt{2.64} \times \sqrt{5.94}}$.

SSC CGL Tier II (26/10/2023)

(a) $\frac{31}{66}$ (b) $\frac{31}{44}$ (c) $\frac{33}{64}$ (d) $\frac{33}{31}$

SSC CPO 2023 Tier - 1

Q.40. Evaluate

$$\frac{0.04}{0.05} \text{ of } \left(3\frac{1}{3} - 2\frac{1}{2}\right) \div \frac{1}{2} \text{ of } 1\frac{1}{4}$$

SSC CPO 03/10/2023 (1st Shift)

(a) 5 (b) 0.3 (c) 3 (d) 0.03

Q.41. If $(48 \div 72 \times 3) - [15 \div 8 \times (40 - 32) - 10] + 2P = 6 \div 2$, then find the value of P

SSC CPO 03/10/2023 (3rd Shift)

(a) 2 (b) 4 (c) 1 (d) 3

Q.42. Simplify $\frac{1 - \frac{3}{7} \div \frac{9}{49} \times \frac{3}{14}}{1 + \frac{3}{5} \div \frac{1}{25} \times \frac{1}{30}}$.

SSC CPO 04/10/2023 (1st Shift)

(a) $\frac{1}{6}$ (b) $\frac{1}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{7}$

Q.43. The value of $1\frac{2}{5} - [3\frac{3}{4} \div \{1\frac{1}{4} \div \frac{1}{2}$

$$(1\frac{1}{2} \times 3\frac{1}{3} \div 1\frac{1}{3})\}$$
 is:

SSC CPO 04/10/2023 (1st Shift)

(a) 1 (b) 3 (c) 2 (d) 0

Q.44. The value of

$$1\frac{1}{2} \div 3\frac{1}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5} \times \frac{1}{5} \times 3\frac{1}{2} - \frac{1}{3} \div 1\frac{3}{4} \times 3\frac{1}{2}$$
 is:

SSC CPO 05/10/2023 (1st Shift)

(a) 38 (b) 33 (c) 36 (d) 40

Q.45. Simplify the given expression.

$$b + 2a - [(3b + a) - (2a + b) + 2a] - a$$

SSC CPO 05/10/2023 (3rd Shift)

(a) -a (b) -b (c) a (d) b

SSC MTS 2023 Tier - 1

Q.46. Simplify:

$$24 \div 6 \times (\frac{3}{4} \text{ of } \frac{1}{3} + \frac{5}{3} \text{ of } \frac{3}{4}) = ?$$

SSC MTS 01/09/2023 (3rd Shift)

(a) 9 (b) 6 (c) 3 (d) 8

Q.47. Simplify the given expression

$$71 - (-3)\{-2 - (8 - 3)\} + 3\{5 + (-2)(-1)\}$$

SSC MTS 04/09/2023 (3rd Shift)

(a) 30 (b) 119 (c) 50 (d) 71

Q.48. Simplify the following. $15 \div 5$ of $3 \times 4 + 8 \div 12$ of $4 \times 12 - 4 \div 5 \times 20$

SSC MTS 06/09/2023 (2nd Shift)

(a) $\frac{20}{3}$ (b) -10 (c) 12 (d) 1

Q.49. The value of

$$\frac{3 \div 3 \times 2 - 6 \div 3 + 7}{5 \div 5 \times 6 \div 6 + 3 + 3} \text{ is equal to:}$$

SSC MTS 08/09/2023 (3rd Shift)

- (a) 1 (b) $\frac{3}{2}$ (c) $1\frac{3}{2}$ (d) 2

Q.50. Evaluate $3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{3}{4}$

$$(1\frac{1}{2} - \frac{1}{2} - 2\frac{1}{2})\}$$

SSC MTS 11/09/2023 (1st Shift)

- (a) $1\frac{21}{38}$ (b) $2\frac{19}{18}$ (c) $2\frac{21}{38}$ (d) $\frac{47}{38}$

Q.51. What is the value of R, if,

$$10(8 + 4R) \div (12 \times 5) - 5 \times 2^3 = 6$$

SSC MTS 11/09/2023 (2nd Shift)

- (a) 86 (b) 68 (c) 76 (d) 67

Q.52. Identify the single rational number that may replace the given expression.

$$[1^2 + 2^2 + 3^2 + 4^2] \times \frac{2}{3 \times 5} \div (\frac{1}{4})^2$$

SSC MTS 11/09/2023 (3rd Shift)

- (a) 16 (b) 64 (c) $\frac{1}{64}$ (d) $\frac{1}{16}$

Q.53. What is the value of

$$\left[\frac{\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5}}{\frac{1}{9} + \frac{1}{9} \times \frac{1}{9}} \right] \times \frac{1}{10} + (1 - \frac{6}{25})$$

SSC MTS 12/09/2023 (2nd Shift)

- (a) 2 (b) 1 (c) 3 (d) 4

Q.54. Simplify the following expression.

$$(\frac{5}{3} + \frac{6}{5} \div \frac{18}{5}) \text{ of } \frac{5}{12} - (\frac{5}{6} \div \frac{5}{12}) \text{ of } \frac{5}{12} + \frac{2}{3} \text{ of } \frac{3}{2}$$

SSC MTS 13/09/2023 (2nd Shift)

- (a) 1 (b) $\frac{5}{6}$ (c) $\frac{6}{5}$ (d) 0

Q.55. The Value of $\{[2^4(20 \times 5 + 10) \div 11] - (2^3 \times 15 \div 20) - (9 \times 16)\}$ is:

SSC MTS 13/09/2023 (3rd Shift)

- (a) 8 (b) 16 (c) 10 (d) 12

Q.56. Simplify: $6\frac{4}{5} \div [4\frac{1}{4} \div \{5\frac{1}{3} \times \frac{3}{8} +$

$$(3\frac{1}{2} + \frac{2}{3}) \times \frac{4}{5}\}$$

SSC MTS 14/09/2023 (1st Shift)

- (a) $8\frac{8}{15}$ (b) $7\frac{8}{25}$ (c) $8\frac{8}{25}$ (d) $7\frac{8}{15}$

SSC CHSL 2023 Tier - 1

Q.57. Simplify the given expression .

$$\frac{4[(17)^3 - (7)^3]}{(17)^2 + (7)^2 + 119}$$

SSC CHSL 02/08/2023 (4th Shift)

- (a) 40 (b) 50 (c) 60 (d) 30

Q.58. Simplify :

$$\frac{(612 + 388)^2 + (612 - 388)^2}{(612 \times 612) + (388 \times 388)} = ?$$

SSC CHSL 04/08/2023 (4th Shift)

- (a) 1 (b) 4 (c) 2 (d) 3

Q.59. If the mathematical symbols in the expression given below are changed from '+' to '÷' '-' to '×', '÷' to '-' and '×' to '+'. then what will be the value of the expression ?

$$87 \times 256 + 16 - 77 \div 1359$$

SSC CHSL 07/08/2023 (3rd Shift)

- (a) 20 (b) -40 (c) 40 (d) -20

Q.60. Simplify the following expression

$$\frac{(0.96 \times 0.96 \times 0.96 + 0.04 \times 0.04 \times 0.04)}{(0.96 \times 0.96 - 0.96 \times 0.04 + 0.04 \times 0.04)}$$

SSC CHSL 07/08/2023 (4th Shift)

- (a) 0 (b) 0.82 (c) 2 (d) 1

Q.61. If $\sqrt{a} = 3b$, then $\frac{a}{b^2}$ is equal to :

SSC CHSL 08/08/2023 (2nd Shift)

- (a) $\frac{1}{6}$ (b) 6 (c) $\frac{1}{9}$ (d) 9

Q.62. Simplify the given expression.

$$\frac{(0.25)^4 + 2 \times (0.25)^2 + 1 - (0.25)^2}{(0.25)^2 + 0.25 + 1}$$

SSC CHSL 08/08/2023 (2nd Shift)

- (a) 0.6755 (b) 0.9025
(c) 0.8125 (d) 0.7835

Q.63. The value of

$$\frac{3 + 8 \times 8 \div 8 \text{ of } 8 + 7 \div 7 \times 2}{5 \div 5 \text{ of } 4 + 3 \times 3 \div 3 - 3 + 2} \text{ is:}$$

SSC CHSL 08/08/2023 (4th Shift)

- (a) $1\frac{3}{5}$ (b) $2\frac{2}{3}$ (c) $1\frac{4}{7}$ (d) $2\frac{3}{7}$

Q.64. Evaluate $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$, given that

$$\sqrt{6} = 2.45 .$$

SSC CHSL 08/08/2023 (4th Shift)

- (a) 7.7 (b) 9.9 (c) 8.8 (d) 6.6

Q.65. Simplify: $[\frac{1}{5} \text{ of } (35 - 5) + 4]^2 - [9$

$$- \{7 + (2 - 3)\} + 5]^2$$

SSC CHSL 09/08/2023 (3rd Shift)

- (a) 100 (b) 36 (c) 25 (d) 64

Q.66. Simplify the given expression.

$$\frac{(80 \times 80 \times 80) + (70 \times 70 \times 70) + (50 \times 50 \times 50) - 840000}{6400 + 4900 + 2500 - 5600 - 3500 - 4000}$$

SSC CHSL 10/08/2023 (1st Shift)

- (a) 100 (b) 200 (c) 400 (d) 300

Q.67. Find the value of the given expression

$$\frac{(4\frac{1}{3} + 3\frac{1}{3} \times 1\frac{4}{5} \div 3\frac{3}{4} \times (1\frac{1}{2} + 1\frac{1}{3}))}{\frac{2}{3} \div \frac{5}{6} \times \frac{2}{3}}$$

SSC CHSL 10/08/2023 (2nd Shift)

$$(a) 11\frac{3}{8} \quad (b) 10\frac{1}{8} \quad (c) 14\frac{3}{8} \quad (d) 16\frac{5}{8}$$

Q.68. Simplify :

$$\frac{(379 + 276)^2 + (379 - 276)^2}{379 \times 379 + 276 \times 276}$$

SSC CHSL 11/08/2023 (2nd Shift)

- (a) 2 (b) 655 (c) 103 (d) 1

Q.69. What is the value of p, if, $25(4 + 3p) \div 5 - 3 \times 8 = 26$?

SSC CHSL 11/08/2023 (2nd Shift)

- (a) 5 (b) 4 (c) 7 (d) 2

Q.70. Simplify the given expression.

$$\frac{(381 + 119)^2 + (381 - 119)^2}{(381)^2 + (119)^2}$$

SSC CHSL 14/08/2023 (1st Shift)

- (a) -1 (b) -2 (c) 1 (d) 2

Q.71. Simplify the given expression

$$\frac{(4.2)^3 - 0.008}{(4.2)^2 + 0.84 + 0.04}$$

SSC CHSL 14/08/2023 (2nd Shift)

- (a) 4 (b) -2 (c) 2 (d) -4

Q.72. The value of

$$(\frac{2 + 7 \times 7 \div 9 \text{ of } 9 + 6 \div 6 \times 2}{4 \div 4 \text{ of } 5 + 7 \times 7 \div 7 - 6 + 4}) \text{ is:}$$

SSC CHSL 17/08/2023 (2nd Shift)

- (a) $\frac{1734}{2407}$ (b) $\frac{1865}{2106}$ (c) $\frac{1857}{2507}$ (d) $\frac{1664}{2305}$

Q.73. Simplify the expression:

$$\frac{143 \times 143 + 143 \times 139 + 139 \times 139}{143 \times 143 \times 143 - 139 \times 139 \times 139}$$

SSC CHSL 17/08/2023 (3rd Shift)

- (a) $\frac{1}{2}$ (b) 282 (c) $\frac{1}{4}$ (d) 4

Q.74. Simplify the given expression and find the value for x = -1.

$$\frac{10x^2 + 5x + 2xy + y}{5x + y}$$

SSC CHSL 17/08/2023 (3rd Shift)

- (a) -1 (b) 0 (c) 1 (d) 2

SSC CGL 2023 Tier - 1

Q.75. Simplify the following expression

$$[\frac{85}{34} \times \frac{1}{18} - \{(\frac{46}{69} \div \frac{27}{135}) - (\frac{86}{129} \times \frac{91}{14})\}] \text{ of } \frac{112}{36}$$

SSC CGL 14/07/2023 (3rd shift)

- (a) $3\frac{1}{5}$ (b) $3\frac{3}{4}$ (c) $3\frac{1}{2}$ (d) $3\frac{1}{4}$

Q.76. Simplify the given expression.

$$18 \div 3 \text{ of } 2 \times 5 + 72 \div 18 \text{ of } 2 \times 3 - 4 \div 8 \times 2$$

SSC CGL 14/07/2023 (4th shift)

- (a) 20 (b) $\frac{20}{3}$ (c) $\frac{25}{3}$ (d) 25

Q.77. Simplify the given expression.

$$y + 2x - [y - (y - x + y) - (x + y) + y] - 2y.$$

SSC CGL 17/07/2023 (1st shift)

(a) -y (b) -2x (c) y (d) 2x

Q.78. Simplify the given expression.

$$\frac{(326 + 222)^2 - (326 - 222)^2}{(326 \times 222)}$$

SSC CGL 18/07/2023 (1st shift)

(a) 1 (b) 4 (c) 3 (d) 2

Q.79. Simplify the following expression.

$$[\sqrt{25} + 12 \div 3 - \{20 + (16 \div 8 \div 16) - (54 \div 18 \text{ of } \frac{1}{2})\}]$$

SSC CGL 18/07/2023 (4th shift)

(a) -13 (b) 12 (c) 22 (d) 0

Q.80. Simplify the following

$$\frac{3a+b}{2} - \frac{a-3b}{3} + 2b$$

SSC CGL 19/07/2023 (1st shift)

$$(a) \frac{5(a+3b)}{6} \quad (b) \frac{7a+3b}{6}$$

$$(c) \frac{7(a+3b)}{6} \quad (d) \frac{a+3b}{6}$$

Q.81. Simplify the given expression.

$$(a - 2b)(b - 3a) + (a + b)(a - 3b) - (b - 3a)(4a - 5b)$$

SSC CGL 21/07/2023 (1st shift)

(a) $8a^2 - 10ab$ (b) $8a^2 + 10ab$
(c) $10a^2 - 14ab$ (d) $10a^2 - 4ab$ **Q.82.** The value of $(3^2)^2 + 3 \times 3 \div 3 - 3$ is :

SSC CGL 21/07/2023 (2nd shift)

(a) 3^2 (b) 9^2 (c) 2^3 (d) 3^0 **Q.83.** If $720 \div 8 + 915 \div 15 - m + 32 \times 5 = 1104 \div 16 \times 111 \div 37$, then the value of m is:

SSC CGL 24/07/2023 (1st shift)

(a) 104 (b) 518 (c) 207 (d) 311

Q.84. Simplify the following expression.

$$(4x + 1)^2 - (4x + 3)(4x - 1)$$

SSC CGL 24/07/2023 (2nd shift)

(a) $(4x + 1)$ (b) 4 (c) $(4x - 3)$ (d) $4x$ **Q.85.** Simplify the following.

$$\frac{0.01 \times 0.01 \times 0.01 + 0.003 \times 0.003 \times 0.003}{0.05 \times 0.05 - 0.015 \times 0.05 + 0.015 \times 0.015}$$

SSC CGL 24/07/2023 (3rd shift)

$$(a) \frac{13}{25} \times 10^3 \quad (b) \frac{13}{15} \times 10^{-3}$$

$$(c) \frac{13}{15} \times 10^3 \quad (d) \frac{13}{25} \times 10^{-3}$$

Q.86. What value of 'y' will satisfy the following equation ?

$$\frac{3y}{1 + \frac{1}{1+y}} = 2$$

SSC CGL 25/07/2023 (1st shift)

(a) $\frac{2}{3}$ (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) $\frac{3}{2}$ **Q.87.** Simplify the expression $\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$,where $x = 2$ and $y = 3$.

SSC CGL 25/07/2023 (4th shift)

$$(a) 2\sqrt{6} - 6 \quad (b) \sqrt{6} - 5$$

$$(c) 5 - 2\sqrt{6} \quad (d) 2\sqrt{6} - 5$$

Q.88. Simplify the given expression

$$\frac{[120 \times 120 \times 120 - 100 \times 100 \times 100]}{[120 \times 120 + 120 \times 100 + 100 \times 100]}$$

SSC CGL 27/07/2023 (2nd shift)

(a) 20 (b) 25 (c) 10 (d) 15

SSC Selection Post (Phase - XI)**Q.89.** Simplify $52 \times 52 - 48 \times 48$.

Higher Secondary 27/06/2023 (Shift - 3)

(a) 386 (b) 400 (c) 424 (d) 326

SSC CHSL 2022 Tier - 2**Q.90.** What is the correct descending order of the following ratios ?

$$\frac{17}{30}, \frac{7}{15}, \frac{27}{50}, \frac{11}{20}$$

SSC CHSL Tier II (26/06/2023)

$$(a) \frac{11}{20} > \frac{17}{30} > \frac{27}{50} > \frac{7}{15}$$

$$(b) \frac{17}{30} > \frac{11}{20} > \frac{27}{50} > \frac{7}{15}$$

$$(c) \frac{17}{30} > \frac{27}{50} > \frac{11}{20} > \frac{7}{15}$$

$$(d) \frac{7}{15} > \frac{11}{20} > \frac{27}{50} > \frac{17}{30}$$

SSC MTS 2022 Tier - 1**Q.91.** Find the value of

$$\sqrt{[(13^2 - 8^2) \times 5 + 4]} :$$

SSC MTS 03/05/2023 (Afternoon)

(a) 22 (b) 24 (c) 26 (d) 23

Q.92. What is the value of

$$\frac{14 + 7 - 12 + 15 - 17 + 2 - 3 + 17}{2 + 7 - 9 + 2 - 4 + 22} ?$$

SSC MTS 03/05/2023 (Afternoon)

$$(a) \frac{23}{20} \quad (b) \frac{17}{20} \quad (c) \frac{19}{20} \quad (d) \frac{33}{20}$$

Q.93. The value of

$$\left(\frac{1}{5}\right) + \left[1\frac{1}{4} - \frac{1}{2}\right] \times \left(\frac{2}{5}\right) \text{ lies between :}$$

SSC MTS 04/05/2023 (Morning)

(a) 2.1 and 2.4 (b) 0.2 and 0.6
(c) 1.5 and 1.8 (d) 0.8 and 1.2**Q.94.** The value of $\left(\frac{1}{2} + \frac{3}{4}\right) \div \left(\frac{1}{5} + \frac{3}{10}\right)$

lies between:

SSC MTS 09/05/2023 (Afternoon)

(a) 2.7 and 3.0 (b) 2.0 and 2.2
(c) 1.5 and 1.8 (d) 2.3 and 2.6**Q.95.** Find the value of $0.5 + 0.55 + 0.555 + 0.5555 :$

SSC MTS 09/05/2023 (Evening)

(a) 1.4405 (b) 1.0555

(c) 1.5555 (d) 2.1605

Q.96. Evaluate: $5 \times 4 + (3 + 4)^2 \div 7 \times 4 :$

SSC MTS 12/05/2023 (Evening)

(a) 47 (b) 46 (c) 48 (d) 49

Q.97. Find the value of

$$\sqrt{[(7^2 + 6^3 + 9^2) \div 2 + 52]}$$

SSC MTS 16/05/2023 (Morning)

(a) 14 (b) 15 (c) 16 (d) 13

Q.98. The value of $74 + 88 \div 8 \times 10$ of $5 - 90 \div 15 + \{44 - (108 \div 12)\}$ is:

SSC MTS 16/05/2023 (Evening)

(a) 700 (b) 653 (c) 770 (d) 605

Q.99. $48 \div 4 + A \times 2 - 8 = 14$, find out the value of A.

SSC MTS 17/05/2023 (Evening)

(a) 10 (b) 5 (c) 14 (d) 8

Q.100. What is the value of x.

$$\text{if } \left(\frac{1}{2}\right) \text{ of } 20 + \left(\frac{1}{4}\right) \text{ of } 28 - \left(\frac{1}{3}\right) \text{ of } x = 10 ?$$

SSC MTS 19/05/2023 (Morning)

(a) 21 (b) 7 (c) 28 (d) 10

Q.101. What is the value of

$$\frac{2.5 \times 2.5 + 0.3 \times 0.3 + 5 \times 0.3}{1.6 \times 1.6 + 0.6 \times 0.6 - 1.2 \times 1.6} ?$$

SSC MTS 19/05/2023 (Evening)

(a) 7.64 (b) 8.64 (c) 2.8 (d) 7.84

Q.102. What is the value of

$$\sqrt{0.52 \times 0.52 + 0.31 \times 0.31 + 0.62 \times 0.52} ?$$

SSC MTS 13/06/2023 (Morning)

(a) 0.63 (b) 0.73 (c) 0.83 (d) 0.93

Q.103. What is the value of A .

$$\text{if } 2 + 4 \div (A + 6) \times 2 = \frac{16}{7} ?$$

SSC MTS 14/06/2023 (Afternoon)

(a) 22 (b) 20 (c) 23 (d) 21

Q.104. What is the value of

$$\frac{1.2 \times 1.2 + 0.2 \times 0.2 + 1.2 \times 0.4}{1.8 \times 1.8 + 0.4 \times 0.4 - 1.8 \times 0.8} ?$$

SSC MTS 15/06/2023 (Morning)

(a) 0.1 (b) 0.25 (c) 0.2 (d) 1

Q.105. What is the value of

$$\sqrt{0.48 \times 0.48 + 0.25 \times 0.25 + 0.5 \times 0.48}$$

SSC MTS 16/06/2023 (Afternoon)

(a) 0.63 (b) 0.93 (c) 0.83 (d) 0.73

Q.106. Find the value of :

$$\frac{49}{63} \div \frac{21}{9} - \frac{85}{510} + \frac{88}{132} \div \frac{66}{11}$$

$$\frac{1}{7} \div \frac{33}{21} + \frac{26}{143} - \frac{1}{209} \div \frac{1}{19}$$

SSC MTS 19/06/2023 (Afternoon)

(a) $\frac{55}{36}$ (b) $\frac{55}{39}$ (c) $\frac{13}{46}$ (d) $\frac{44}{13}$

SSC CHSL 2022 Tier - 1**Q.107.** What is the value of

$$\frac{\frac{5}{2} \div \frac{2}{5} \times \frac{1}{5} - \frac{1}{5} \div \frac{3}{5} \times \frac{5}{2}}{\frac{2}{5} \div \frac{1}{5} - \frac{1}{2} \times \frac{5}{2} \div \frac{25}{4} + \frac{1}{2} \div \frac{2}{5} \times \frac{1}{5}}$$

SSC CHSL 09/03/2023 (1st Shift)

- (a)
- $\frac{31}{127}$
- (b)
- $\frac{32}{127}$
- (c)
- $\frac{28}{125}$
- (d)
- $\frac{25}{123}$

Q.108. If $A = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))}$ and $B = \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$ then what is the value of $(A + B)$?

SSC CHSL 09/03/2023 (4th Shift)

- (a)
- $\frac{776}{225}$
- (b)
- $\frac{424}{225}$
- (c)
- $\frac{772}{225}$
- (d)
- $\frac{401}{225}$

Q.109. What is the difference of the largest and smallest of the given fractions ?

$$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{13}$$

SSC CHSL 13/03/2023 (1st Shift)

- (a)
- $\frac{17}{56}$
- (b)
- $\frac{19}{56}$
- (c)
- $\frac{1}{7}$
- (d)
- $\frac{23}{56}$

Q.110. What is the value of

$$\frac{(.91)^3 + (.09)^3}{[(.91)^2 - .0819 + (.09)^2]}$$

SSC CHSL 15/03/2023 (1st Shift)

- (a) 1 (b) 5 (c) 4 (d) 6

Q.111. If $\frac{1}{4}$ of $\frac{4}{5} + \frac{9}{2}$ of $\frac{17}{3} - \frac{3}{10} = p$,then what is the value of p ?

SSC CHSL 20/03/2023 (1st Shift)

- (a)
- $\frac{127}{5}$
- (b)
- $\frac{177}{5}$
- (c)
- $\frac{255}{2}$
- (d)
- $\frac{51}{2}$

Q.112. If $A = \frac{18 \div 9 \times 4}{15 \div 3 \times 5}$ and $B = \frac{18 \div 36 \times 3 + 5 \times 1 - 6}{18 \div 6 \times 20 - 3 \times 4 + 12}$ then what is the value of $(A + B)$?

SSC CHSL 21/03/2023 (3rd Shift)

- (a)
- $\frac{137}{600}$
- (b)
- $\frac{23}{100}$
- (c)
- $\frac{187}{600}$
- (d)
- $\frac{197}{600}$

SSC CGL 2022 Tier - 2**Q.113.** Find the value of the given expression.

$$\sqrt{20 - \sqrt{20 - \sqrt{20 - \sqrt{20 - \dots}}}}$$

SSC CGL Tier II (03/03/2023)

- (a) 4 (b) 6 (c) 5 (d) 2

Q.114. Simplify the following.

$$\frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}}{\sqrt{16 + 19.25 \times 4^2}}$$

SSC CGL Tier II (06/03/2023)

- (a)
- $\frac{7}{18}$
- (b)
- $\frac{2}{9}$
- (c)
- $\frac{5}{18}$
- (d)
- $\frac{1}{9}$

SSC CGL 2022 Tier - 1**Q.115.** Simplify $x^9 \times x^5 \times x^{-4} \times x^0 \times x^{-6}$

SSC CGL 01/12/2022 (2nd Shift)

- (a)
- x^4
- (b)
- x^{-4}
- (c)
- x^{-6}
- (d)
- x^6

Q.116. What is the value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \dots$

SSC CGL 05/12/2022 (3rd Shift)

- (a) 5050 (b) 4985 (c) 4995 (d) 4950

Q.117. Simplify the given expression.

$$(5x + 8y) (25x^2 + 64y^2 - 40xy)$$

SSC CGL 07/12/2022 (3rd Shift)

- (a)
- $125x^3 - 512y^3$
- (b)
- $25x^3 + 64y^3$
-
- (c)
- $125x^3 + 512y^3$
- (d)
- $25x^3 - 64y^3$

Q.118. Simplify $\sqrt{36x^2 - 108x + 81}$.

SSC CGL 07/12/2022 (3rd Shift)

- (a)
- $6x - 9$
- (b)
- $2x - 9$
- (c)
- $5x - 9$
- (d)
- $3x - 3$

SSC CPO 2022 Tier - 1**Q.119.** The value of

$$\frac{\sqrt[3]{-2744} \times \sqrt[3]{-216}}{\sqrt[3]{\frac{64}{729}}}$$
 is

SSC CPO 09/11/2022 (Morning)

- (a) 164 (b) 152 (c) 189 (d) 156

Q.120. Simplify the following. $81^{\frac{3}{4}} + [(20$

$$\div 5 \text{ of } 3 \times 6) + \{(8 \div 24 \text{ of } 3) \times 4\} - 10 \div 5] - \left(\frac{1}{32}\right)^{-\frac{2}{5}}$$

SSC CPO 09/11/2022 (Morning)

- (a)
- $24\frac{1}{4}$
- (b)
- $21\frac{1}{9}$
- (c)
- $27\frac{4}{5}$
- (d)
- $29\frac{4}{9}$

Q.121. What is the value of 'a' in the below equation ?

$$\{(5 \times 5 \times 5 \times 5 \times 5 \times 5) \times (5 \times 5 \times 5 \times 5 \times 5)\} \div (5 \times 5) = (625)^a$$

SSC CPO 09/11/2022 (Afternoon)

- (a) 14 (b) 17 (c) 16 (d) 15

Q.122. Solve the following equation.

$$(7^3 - 7^2 + 6) \times (6^3 - 2^3 - 3^2 + 1) \div (21^2 + 20^2 - 41) = \underline{\hspace{2cm}}$$

SSC CPO 10/11/2022 (Afternoon)

- (a) 75 (b) 40 (c) 50 (d) 25

Q.123. What is the positive value of the following expression ?

$$\sqrt{36 \div 15 \text{ of } 2 \text{ of } [25 \times 4 \div 4 \text{ of } \{29 + (12 - 11)\}]}$$

SSC CPO 11/11/2022 (Afternoon)

- (a)
- $1\frac{5}{6}$
- (b)
- $1\frac{1}{5}$
- (c)
- $2\frac{4}{5}$
- (d)
- $2\frac{3}{5}$

SSC CGL 2021 Tier - 2**Q.124.** What is the value of

$$99\frac{11}{99} + 99\frac{13}{99} + 99\frac{15}{99} + \dots + 99\frac{67}{99} ?$$

SSC CGL Tier II (08/08/2022)

- (a)
- $\frac{94220}{33}$
- (b)
- $\frac{95120}{33}$
-
- (c)
- $\frac{97120}{33}$
- (d)
- $\frac{96220}{33}$

Q.125. What is the simplified value of

$$\frac{(x + y + z)(xy + yz + zx) - xyz}{(x + y)(y + z)(z + x)} ?$$

SSC CGL Tier II (08/08/2022)

- (a) y (b) z (c) 1 (d) x

Q.126. What is the value of

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}} + \frac{\sqrt{0.2916}}{\sqrt{116.64}} + \frac{\sqrt{0.0036}}{\sqrt{0.36}} ?$$

SSC CGL Tier II (08/08/2022)

- (a)
- $\frac{26}{5}$
- (b)
- $\frac{103}{20}$
- (c)
- $\frac{27}{5}$
- (d)
- $\frac{101}{20}$

Q.127. If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and $B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is $A \times B$?

SSC CGL Tier II (08/08/2022)

- (a)
- 7×10^{-7}
- (b)
- $\left(\frac{7}{4}\right) \times 10^{-8}$
-
- (c)
- $\frac{3}{5} \times 10^{-7}$
- (d)
- $\left(\frac{7}{3}\right) \times 10^{-7}$

Q.128. If $A = 0.3\overline{12}$, $B = 0.4\overline{15}$ & $C = 0.30\overline{9}$, then what is the value of $A + B + C$?

SSC CGL Tier II (08/08/2022)

- (a)
- $\frac{1141}{1100}$
- (b)
- $\frac{1097}{1100}$
- (c)
- $\frac{1211}{1100}$
- (d)
- $\frac{1043}{1100}$

SSC Selection Post (Phase - X)**Q.129.** If $\sqrt{5} = 2.24$, then find the value of

$$\sqrt{\frac{\sqrt{5} - 1}{\sqrt{5} + 1}}$$

Graduate Level 04/08/2022 (Shift - 4)

- (a) 0.62 (b) 0.59 (c) 0.68 (d) 0.55

Q.130. Find the value of : $13^3 + (-8)^3 + (-5^3)$

Higher Secondary 01/08/2022 (Shift - 3)

- (a) 1052 (b) 1152 (c) 1562 (d) 1560

SSC MTS 2021 Tier - 1**Q.131.** Select the number that will come in place of the question mark(?) in the mathematical statement.

$$(0.064)^{123} \div (0.16)^{47} \times (0.4)^{34} \times (0.4)^{29} = (0.4)^?$$

SSC MTS 08/07/2022 (Afternoon)

- (a) 350 (b) 320 (c) 338 (d) 341

Q.132. If $\frac{2x+3}{18 \div 3 \times 4 + 2} = \frac{12 \times 3 + 5}{2 \times 4 \div 2}$,

then what is the value of x ?

SSC MTS 11/07/2022 (Evening)

(a) 132.25 (b) 131.75 (c) 134.35 (d) 120.5

Q.133. If $A = \frac{2}{3} + \frac{4}{7}$ of $\frac{14}{12}$ and $B = \frac{5}{12}$

of $\frac{7}{15} \times \frac{36}{21}$, then what is the value of A + B ?

SSC MTS 18/07/2022 (Morning)

(a) $\frac{5}{3}$ (b) 2 (c) $\frac{4}{3}$ (d) $\frac{2}{3}$

Q.134. What is the value of $\frac{\frac{5}{6} - \frac{5}{11} \text{ of } \left[\frac{3}{7} + \frac{8}{12} \div \frac{6}{24} + \frac{5}{7} \right]}{\frac{3}{12} \times \frac{6}{5}}$?

SSC MTS 22/07/2022 (Afternoon)

(a) $\frac{-1891}{691}$ (b) $\frac{2075}{693}$
(c) $\frac{-1894}{891}$ (d) $\frac{-2075}{693}$

Q.135. If $P = 7 \times 4 - 3$, $Q = 8 \div 4 + 2$, and $R = 14 \div 21 \times 9$, then what is the value of $P \times Q + R$?

SSC MTS 25/07/2022 (Morning)

(a) 96 (b) 112 (c) 108 (d) 106

SSC CHSL 2021 Tier - 1

Q.136. Simplify the expression

$$(36p^2 + 49q^2)(6p + 7q)(6p - 7q)$$

SSC CHSL 24/05/2022 (Morning)

(a) $1296p^4 + 2401q^4$ (b) $36p^4 - 49q^4$
(c) $1296p^4 - 2401q^4$ (d) $36p^4 + 49q^4$

Q.137. Simplify :

$$3p - [3p - \overline{p + q} - \{3p - (\overline{p - q - p})\}]$$

SSC CHSL 24/05/2022 (Afternoon)

(a) $3p - q$ (b) $p + 2q$
(c) $3p + 2q$ (d) $2p + 2q$

Q.138. Simplify $[7\frac{1}{2} \div \{1\frac{1}{4} - \frac{1}{2}(2\frac{1}{2} - \frac{1}{4} - \frac{1}{6})\}] \div (\frac{1}{2} \text{ of } 8\frac{1}{3}) = ?$

SSC CHSL 25/05/2022 (Afternoon)

(a) 26 (b) 36 (c) 43.2 (d) 23.6

Q.139. $\{(28 - 3^3) + 99\} + 136 \div 4 \times 5 - (44 - 2^2) = ?$

SSC CHSL 30/05/2022 (Afternoon)

(a) 250 (b) 200 (c) 230 (d) 270

Q.140. Find the value of

$$\left(\frac{9^3 + 9^4 + 9^5 + 9^6 + 9^7}{9^1 + 9^2 + 9^3 + 9^4 + 9^5} \right)^{\frac{1}{2}} = ?$$

SSC CHSL 01/06/2022 (Morning)

(a) 9 (b) 3 (c) 729 (d) 81

Q.141. Simplify the following expression

$$3\frac{6}{7} - \{5\frac{1}{7} - (2\frac{3}{7} - 3\frac{5}{7} - 4)\}$$

SSC CHSL 01/06/2022 (Afternoon)

(a) $\frac{34}{7}$ (b) $\frac{10}{7}$ (c) $\frac{20}{7}$ (d) $\frac{44}{7}$

Q.142. If a, b, c are positive real numbers, what is the value of

$$\sqrt{a^{-1}b} \times \sqrt{b^{-1}c} \times \sqrt{c^{-1}a} ?$$

SSC CHSL 03/06/2022 (Evening)

(a) 2 (b) 3 (c) 1 (d) 4

Q.143. Simplify the following expression.

$$(2.19)^2 + 2(2.19)(3.81) + (3.81)^2$$

SSC CHSL 06/06/2022 (Morning)

(a) 216 (b) 12 (c) 36 (d) 6

Q.144. The value of

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3}$$

is equal to

SSC CHSL 06/06/2022 (Morning)

(a) $ab + bc + ca$ (b) 0

(c) 1 (d) $\frac{1}{(a+b)(b+c)(c+a)}$

Q.145. Simplify the following expression.

$$\frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca}$$

Where; a = 2.01, b = 39.75, c = 28.24

SSC CHSL 06/06/2022 (Afternoon)

(a) 140 (b) 50 (c) 70 (d) 100

Q.146. Find the value of

$$\sqrt{2 + \sqrt{3}} + \sqrt{2 - \sqrt{3}}.$$

SSC CHSL 06/06/2022 (Evening)

(a) $\sqrt{6}$ (b) $2\sqrt{3}$ (c) $2\sqrt{2}$ (d) 6

Q.147. Simplify :

$$\frac{(r-s)^3 + (s-t)^3 + (t-r)^3}{6(r-s)(s-t)(t-r)}$$

SSC CHSL 09/06/2022 (Afternoon)

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

Q.148. Simplify: $(a^{-1} + b^{-1}) \div (a^{-3} + b^{-3})$

SSC CHSL 09/06/2022 (Evening)

(a) $\frac{ab}{a^2 - ab + b^2}$ (b) $\frac{a^3 b^3}{a^2 - ab + b^2}$
(c) $\frac{a^2 b^2}{a^2 + ab + b^2}$ (d) $\frac{a^2 b^2}{a^2 - ab + b^2}$

Q.149. Simplify $(5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^?$

SSC CHSL 10/06/2022 (Morning)

(a) 15 (b) 21 (c) 13 (d) 14

SSC CGL 2021 Tier - 1

Q.150. Simplify the following expressions:

$$\frac{(a^2 - 4b^2)^3 + 64(b^2 - 4c^2)^3 + (16c^2 - a^2)^3}{(a - 2b)^3 + (2b - 4c)^3 + (4c - a)^3}$$

SSC CGL 11/04/2022 (Afternoon)

(a) $-(a - 2b)(b + 2c)(4c + a)$
(b) $2(a + 2b)(b + 2c)(4c + a)$
(c) $(a + 2b)(b + 2c)(4c + a)$
(d) $4(a + 2b)(b + 2c)(4c + a)$

Q.151. If $5\sqrt{3} + \sqrt{75} = 17.32$, then the value of $14\sqrt{3} + \sqrt{108}$ is:

SSC CGL 20/04/2022 (Evening)

(a) 32.46 (b) 35.64 (c) 34.64 (d) 33.86

SSC CGL 2020 Tier - 2

Q.152. If $847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d$, then the value of $ab - cd$ is:

SSC CGL Tier II (29/01/2022)

(a) 5 (b) 1 (c) 4 (d) 7

Q.153. The value of

$$\frac{1}{4 - \sqrt{15}} - \frac{1}{\sqrt{15} - \sqrt{14}} + \frac{1}{\sqrt{14} - \sqrt{13}} - \frac{1}{\sqrt{13} - \sqrt{12}} + \frac{1}{\sqrt{12} - \sqrt{11}} - \frac{1}{\sqrt{11} - \sqrt{10}} + \frac{1}{\sqrt{10} - 3} - \frac{1}{3 - \sqrt{8}}$$

is:

SSC CGL Tier II (29/01/2022)

(a) $4 - 2\sqrt{2}$ (b) $2 + 2\sqrt{2}$

(c) $4 + 2\sqrt{2}$ (d) $2 - 2\sqrt{2}$

Q.154. If $\frac{\sqrt{38 - 5\sqrt{3}}}{\sqrt{26 + 7\sqrt{3}}} = \frac{a + b\sqrt{3}}{23}$, $b > 0$,

then the value of $(b - a)$ is:

SSC CGL Tier II (03/02/2022)

(a) 7 (b) 18 (c) 29 (d) 11

SSC MTS 2020 Tier - 1

Q.155. The value of $\frac{3}{8} + \frac{1}{7} \times 1\frac{3}{4} +$

$$(\frac{2}{3} \div \frac{3}{10} \text{ of } \frac{4}{9}) - (3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3}) \times \frac{15}{8} \text{ is:}$$

SSC MTS 05/10/2021 (Afternoon)

(a) $5\frac{1}{2}$ (b) $10\frac{1}{2}$ (c) $10\frac{1}{4}$ (d) $5\frac{3}{8}$

Q.156. What is the simplified value of

$$\{4 - \frac{2}{1 + \frac{2}{1 - \frac{1}{2 + \frac{3}{4}}}}\} \div 1\frac{5}{12} \text{ of } \frac{72}{145} -$$

$$(4 + 3 \div 0.5 - 1)$$

SSC MTS 20/10/2021 (Afternoon)

(a) -4 (b) 1 (c) 3 (d) -2

Q.157. The value of $4\frac{2}{5} \div \{(1\frac{1}{2} - 3\frac{1}{5})$

$$\div 3\frac{2}{5} + (2\frac{1}{5} \div 1\frac{1}{2} + 4\frac{2}{5}) + \frac{1}{2} = ?$$

SSC MTS 22/10/2021 (Evening)

(a) $\frac{4}{7}$ (b) $\frac{3}{7}$ (c) $\frac{3}{5}$ (d) $\frac{3}{4}$

SSC CGL 2020 Tier - 1

Q.158. The value of $3\frac{5}{6} + [3\frac{2}{3} + \{ \frac{15}{4} (5\frac{4}{5} \div 14\frac{1}{2}) \}]$ is equal to:

SSC CGL 16/08/2021 (Afternoon)

(a) 9 (b) 6 (c) 7 (d) 8

Q.159. Simplify the following expression:

$$(\frac{3}{4} - \frac{1}{4} \div \frac{1}{4} \text{ of } \frac{2}{5}) \div (\frac{3}{4} \div \frac{2}{3} \text{ of } \frac{3}{5})$$

SSC CGL 24/08/2021 (Evening)

(a) $\frac{14}{75}$ (b) $-\frac{70}{27}$ (c) $-\frac{14}{15}$ (d) $\frac{32}{75}$

SSC CHSL 2020 Tier - 1**Q.160.** Simplify the following expression.

$$(2x - 3y)^3 - 18xy(2x - 3y)$$

SSC CHSL 12/04/2021 (Morning)

(a) $8x^3 - 27y^3 - 36x^2y - 54xy^2$

(b) $8x^3 - 27y^3 - 72x^2y + 108xy^2$

(c) $8x^3 - 27y^3$

(d) $8x^3 + 108xy^2 - 72x^2y$

Q.161. The value of

$$364 - [142 - \{75 + (38 - \frac{5}{4} + \frac{14}{8})\}]$$
 is:

SSC CHSL 11/08/2021 (Evening)

(a) 332 (b) 330 (c) 331 (d) 334

SSC CGL 2019 Tier - 2**Q.162.** The value of

$$(2.\bar{4} \times 0.\bar{6} \times 3 \times 0.\bar{16}) \times [0.\bar{27} \times (0.\bar{83} \div 0.\bar{16})]$$
 is:

SSC CGL Tier II (15/10/2020)

(a) $0.\bar{814}$ (b) $0.\bar{11}$ (c) $1.\bar{1}$ (d) $1.\bar{36}$

Q.163. The expression

$$\frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + \sqrt{20} + \sqrt{40} - \sqrt{5} - \sqrt{80}}$$
 is equal to:

SSC CGL Tier II (15/10/2020)

(a) $5 + 2\sqrt{2}$ (b) $5 - 2\sqrt{5}$

(c) $5(3 + 2\sqrt{2})$ (d) $10(3 + 2\sqrt{5})$

Q.164. If $\sqrt{11 - 3\sqrt{8}} = a + b\sqrt{2}$, then what is the value of $(2a + 3b)$?

SSC CGL Tier II (16/11/2020)

(a) 5 (b) 7 (c) 9 (d) 3

Q.165. The value of

$$\frac{7 + 3\sqrt{5}}{3 + \sqrt{5}} - \frac{7 - 3\sqrt{5}}{3 - \sqrt{5}}$$
 lies between:

SSC CGL Tier II (16/11/2020)

(a) 2 and 2.5

(b) 3 and 3.5

(c) 1.5 and 2

(d) 2.5 and 3

Q.166. If $x = \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7} + 4\sqrt{3}}}$ where $x > 0$, then the value of x is equal to:
SSC CGL Tier II (18/11/2020)

(a) 2 (b) 3 (c) 4 (d) 1

Q.167. The value of

$$5 - \frac{8 + 2\sqrt{15}}{4} - \frac{1}{8 + 2\sqrt{15}}$$
 is equal to:

SSC CGL Tier II (18/11/2020)

(a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

SSC CHSL 2019 Tier - 1**Q.168.** Find the value of x .

$$\sqrt[3]{15625} - \sqrt{x} = 4$$

SSC CHSL 14/10/2020 (Evening)

(a) 625 (b) 343 (c) 441 (d) 81

Q.169. If x is the square of the number

$$\text{when } (\frac{2}{5} \text{ of } 6\frac{1}{4} \div \frac{3}{7}) \text{ of } 1\frac{2}{7} \text{ is divided}$$

$$\text{by } 11\frac{1}{4}, \text{ then the value of } 81x \text{ is:}$$

SSC CHSL 18/03/2020 (Morning)

(a) 36 (b) 16 (c) 9 (d) 4

SSC CGL 2019 Tier - 1**Q.170.** The value of

$$\frac{5\frac{1}{2} \div 3\frac{2}{3} \text{ of } \frac{1}{4} + (5\frac{1}{9} - 7\frac{7}{8} \div 9\frac{9}{20}) \times \frac{9}{11}}{5 \div 5 \text{ of } \frac{1}{10} - 10 \times 10 \div 20}$$

Is:

SSC CGL 04/03/2020 (Afternoon)

(a) $1\frac{4}{5}$ (b) $1\frac{9}{10}$ (c) $3\frac{4}{5}$ (d) $9\frac{1}{2}$

SSC CPO 2019 Tier - 1**Q.171.** The value of

$$\frac{\frac{1}{3} + [4\frac{3}{4} - (3\frac{1}{6} - 2\frac{1}{3})]}{(\frac{1}{5} \text{ of } \frac{1}{5} \div \frac{1}{5}) \div (\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5})}$$

lies between

SSC CPO 11/12/ 2019 (Evening)

(a) 10.2 and 10.8 (b) 4.2 and 4.4

(c) 8.2 and 8.8 (d) 0.4 and 0.9

Q.172. Let $x = (\frac{\sqrt{1875}}{\sqrt{3888}} \div \frac{\sqrt{1200}}{\sqrt{768}}) \times$

$$\frac{\sqrt{175}}{\sqrt{1792}}$$
 Then $\sqrt{x}$ is equal to:

SSC CPO 12/12/2019 (Morning)

(a) $\frac{7}{12}$ (b) $\frac{4}{9}$ (c) $\frac{5}{12}$ (d) $\frac{5}{9}$

Q.173. A student was asked to find the value of

$$(2\frac{1}{3} + 2\frac{1}{2} - \frac{1}{6}) \div 2\frac{1}{3} \times 5\frac{2}{3} \div 1\frac{2}{3} \text{ of } 4\frac{1}{4}$$

$$3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3} + 5\frac{1}{3} \times \frac{3}{4} \div 2\frac{2}{3}$$

His answer was $\frac{6}{7}$. What is the difference between the correct answer and his answer?

SSC CPO 13/12/2019 (Morning)

(a) $\frac{9}{14}$ (b) $\frac{5}{14}$ (c) $\frac{11}{49}$ (d) $\frac{6}{49}$

SSC CGL 2018 Tier - 2**Q.174.** If $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$, then what will be the value of k ?

SSC CGL Tier II (11/09/2019)

(a) $(\sqrt{2} + \sqrt{5} + \sqrt{3})$

(b) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$

(c) $(\sqrt{2} + \sqrt{5} - \sqrt{3})(2 + \sqrt{5})$

(d) $(\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{5})$

Q.175. The value of $(1\frac{1}{3} \div 2\frac{6}{7} \text{ of } 5\frac{3}{5})$

$$\div (6\frac{2}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3}) \times (\frac{3}{4} \times 2\frac{2}{3} \div$$

$$\frac{5}{9} \text{ of } 1\frac{1}{5}) = 1 + k, \text{ where } k \text{ lies}$$

between:

SSC CGL Tier II (12/09/2019)

(a) -0.07 and -0.06 (b) -0.08 and -0.07

(c) -0.06 and -0.05 (d) -0.05 and -0.04

Q.176. The expression

$$\sqrt{10} + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})$$
 is equal to

SSC CGL Tier II (12/09/2019)

(a) $\sqrt{3} + \sqrt{2} - \sqrt{5}$ (b) $\sqrt{3} - \sqrt{2} - \sqrt{5}$

(c) $\sqrt{3} - \sqrt{2} + \sqrt{5}$ (d) $\sqrt{2} - \sqrt{3} - \sqrt{5}$

Q.177. The value of $0.5\bar{6} - 0.7\bar{23} + 0.3\bar{9} \times 0.\bar{7}$ is:

SSC CGL Tier II (12/09/2019)

(a) $0.15\bar{4}$ (b) $0.15\bar{4}$

(c) $0.15\bar{8}$ (d) $0.15\bar{8}$

Q.178. The value of

$$\sqrt{28 + 10\sqrt{3}} - \sqrt{7 - 4\sqrt{3}}$$
 is closest to:

SSC CGL Tier II (13/09/2019)

(a) 7.2 (b) 6.1 (c) 6.5 (d) 5.8

Q.179. The value of

$$\frac{2\sqrt{10}}{\sqrt{5} + \sqrt{2} - \sqrt{7}} - \sqrt{\frac{\sqrt{5} - 2}{\sqrt{5} + 2}} - \frac{3}{\sqrt{7} - 2}$$
 is:

SSC CGL Tier II (13/09/2019)

(a) $2 + \sqrt{2}$ (b) $2\sqrt{5}$ (c) $\sqrt{2}$ (d) $\sqrt{7}$

Q.180. A student was asked to find the value of $9\frac{4}{9} \div 11\frac{1}{3} \text{ of } \frac{1}{6} + (1\frac{1}{3} \times 1\frac{4}{5} \div \frac{3}{5}) \times 2\frac{1}{6} \text{ of } \frac{2}{3} \div \frac{4}{3} \text{ of } \frac{2}{3}$ His answer was $19\frac{1}{4}$. What is the difference

between his answer and the correct answer ?

SSC CGL Tier II (13/09/2019)

(a) $7\frac{3}{4}$ (b) $6\frac{2}{3}$ (c) $7\frac{1}{2}$ (d) $6\frac{1}{3}$

SSC MTS 2019 Tier - 1

Q.181. The value of $99\frac{95}{99} \times 99 - 95$ is:

SSC MTS 09/08/2019 (Morning)

(a) 9897 (b) 9993 (c) 9999 (d) 9801

SSC CHSL 2018 Tier - 1

Q.182. What is the simplified value of $5 \div 10$ of $10 \times 4 + 4 \div 4$ of 4

$\times 10 - (10 - 4) \div 16 \times 4$

SSC CHSL 03/07/2019 (Evening)

(a) 1.2 (b) 2.5 (c) 21 (d) 58.5

Q.183. The simplified value of

$$\left(3\frac{1}{5} - \frac{3}{5}\right) \div \frac{8}{5}$$

is:

$$1\frac{1}{7} \div \left\{\frac{6}{7} - \left(\frac{1}{7} \div \frac{1}{5}\right)\right\}$$

SSC CHSL 05/07/2019 (Afternoon)

(a) $\frac{13}{64}$ (b) $\frac{13}{16}$ (c) $\frac{13}{8}$ (d) $\frac{13}{7}$

SSC CGL 2018 Tier - 1

Q.184. Value of $\frac{8}{9}$ of $(5\frac{1}{4} \div 2\frac{1}{3}$ of 4)

$\div (8 \div \frac{2}{3}$ of $\frac{4}{5})$ of $(8 \times \frac{2}{3} \div \frac{4}{5})$ is

SSC CGL 07/06/2019 (Morning)

(a) $1\frac{1}{8}$ (b) $\frac{4}{15}$ (c) $\frac{1}{200}$ (d) $\frac{1}{100}$

Answer Key :-

1.(a)	2.(b)	3.(a)	4.(a)
5.(c)	6.(b)	7.(d)	8.(a)
9.(b)	10.(d)	11.(c)	12.(c)
13.(d)	14.(d)	15.(d)	16.(a)
17.(c)	18.(a)	19.(c)	20.(a)
21.(b)	22.(d)	23.(a)	24.(d)
25.(d)	26.(d)	27.(a)	28.(b)
29.(d)	30.(c)	31.(d)	32.(d)
33.(d)	34.(c)	35.(c)	36.(b)
37.(b)	38.(b)	39.(a)	40.(c)
41.(d)	42.(c)	43.(a)	44.(c)
45.(b)	46.(b)	47.(d)	48.(b)
49.(a)	50.(c)	51.(d)	52.(b)
53.(d)	54.(a)	55.(c)	56.(a)
57.(a)	58.(c)	59.(b)	60.(d)
61.(d)	62.(c)	63.(b)	64.(b)
65.(b)	66.(b)	67.(d)	68.(a)

69.(d)	70.(d)	71.(a)	72.(b)
73.(c)	74.(a)	75.(d)	76.(a)
77.(d)	78.(b)	79.(a)	80.(c)
81.(c)	82.(b)	83.(a)	84.(b)
85.(d)	86.(c)	87.(d)	88.(a)
89.(b)	90.(b)	91.(d)	92.(a)
93.(b)	94.(d)	95.(d)	96.(c)
97.(b)	98.(b)	99.(b)	100.(a)
101.(d)	102.(c)	103.(a)	104.(d)
105.(d)	106.(a)	107.(d)	108.(a)
109.(b)	110.(a)	111.(a)	112.(d)
113.(a)	114.(b)	115.(a)	116.(c)
117.(c)	118.(a)	119.(c)	120.(d)
121.(b)	122.(a)	123.(b)	124.(b)
125.(c)	126.(b)	127.(c)	128.(a)
129.(a)	130.(d)	131.(c)	132.(b)
133.(a)	134.(d)	135.(d)	136.(c)
137.(d)	138.(c)	139.(c)	140.(a)
141.(b)	142.(c)	143.(c)	144.(d)
145.(c)	146.(a)	147.(b)	148.(d)
149.(c)	150.(b)	151.(c)	152.(a)
153.(a)	154.(c)	155.(d)	156.(a)
157.(d)	158.(a)	159.(c)	160.(b)
161.(a)	162.(c)	163.(c)	164.(d)
165.(a)	166.(a)	167.(b)	168.(c)
169.(a)	170.(b)	171.(b)	172.(c)
173.(d)	174.(b)	175.(a)	176.(a)
177.(a)	178.(c)	179.(c)	180.(a)
181.(d)	182.(a)	183.(a)	184.(c)

Solutions :-

Sol.1.(a)

$$= \frac{(x^2-9) \times (9x^2-1)}{\left(x-\frac{1}{3}\right) \times \left(1-\frac{x}{3}\right)} \times \frac{9}{\left(x+\frac{1}{3}\right)\left(1+\frac{x}{3}\right)}$$

$$= \frac{(x+3)(x-3)(3x+1)(3x-1)}{\frac{(3x-1)(3-x)}{9}} \times \frac{9}{(3x+1)(3+x)} = -729$$

Sol.2.(b)

$$A = \frac{58^2 - 25^2}{46^2 - 37^2} = \frac{(58-25)(58+25)}{(46-37)(46+37)}$$

$$= \frac{33 \times 83}{9 \times 83} = \frac{11}{3}$$

$$B = \frac{26^2 - 15^2}{56^2 - 15^2} = \frac{(26-15)(26+15)}{(56-15)(56+15)}$$

$$= \frac{11 \times 41}{41 \times 71} = \frac{11}{71}$$

$$\text{Then, } \frac{1}{B} - \frac{20}{A} = \frac{71}{11} - \frac{20 \times 3}{11} = \frac{11}{11} = 1$$

Sol.3.(a)

$$\Rightarrow \left[\left(2\frac{7}{9}\right)^2\right]^{\frac{3}{5}} = \left[\left(\frac{5}{3}\right)^2\right]^{\frac{5}{2}} = \left(\frac{5}{3}\right)^3$$

$$\Rightarrow \left[\left(1\frac{2}{3}\right)^5\right]^{\frac{3}{5}} = \left(\frac{5}{3}\right)^3$$

$\therefore$ both the expressions are equal

Sol.4.(a)

$$\frac{24\sqrt{45} - 16\sqrt{20} + \sqrt{245} - 47\sqrt{5}}{3\frac{1}{5} + 1\frac{2}{5} - 2\frac{1}{6} - 2\frac{1}{5} - 5\frac{2}{5} + 3\frac{1}{6}}$$

$$= \frac{72\sqrt{5} - 32\sqrt{5} + 7\sqrt{5} - 47\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}}$$

$$= \frac{79\sqrt{5} - 79\sqrt{5}}{\frac{16}{5} + \frac{7}{5} - \frac{13}{6} - \frac{11}{5} - \frac{27}{5} + \frac{19}{6}} = 0$$

Sol.5.(c)

$$\frac{(3.321)^3 + (2.681)^3 + (1.245)^3 - 3 \times 3.321 \times 2.681 \times 1.245}{(3.321)^2 + (2.681)^2 + (1.245)^2 - (3.321 \times 2.681) - (2.681 \times 1.245) - (1.245 \times 3.321)}$$

$$\frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)$$

Here, $3.321 + 2.681 + 1.245 = 7.247$

Sol.6.(b)

$$0.01 + 0.68 \times 0.68 + 0.22 \times 0.22 + 0.044 + 0.136 + 0.44 \times 0.68$$

$$= (0.68 + 0.22)^2 + 0.01 + 0.044 + 0.136$$

$$0.81 + 0.19 = 1$$

$$\text{Sol.7.(d)} \quad (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\frac{0.09 \times 0.09 + 0.04 \times 0.04 + 0.16 \times 0.16 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.3 \times 0.3 + 0.2 \times 0.2 + 0.4 \times 0.4}$$

$$\frac{(0.09)^2 + (0.04)^2 + (0.16)^2 + 2 \times 0.09 \times 0.04 + 2 \times 0.04 \times 0.16 + 2 \times 0.09 \times 0.16}{0.09 + 0.04 + 0.16}$$

$$= \frac{(0.09 + 0.04 + 0.16)^2}{0.09 + 0.04 + 0.16}$$

$$= 0.09 + 0.04 + 0.16 = 0.29$$

$$\text{Sol.8.(a)} \quad \frac{4^{a+4} - 5 \times 4^{a+2}}{15 \times 4^a - 2^2 \times 4^a}$$

$$= \frac{4^a(4^4 - 5 \times 4^2)}{4^a(15 - 2^2)} = \frac{(256 - 80)}{11} = 16$$

Sol.9.(b)

$$(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)$$

Multiply and divide by $(2^2 - 1)$ in the given equation, we get

$$\frac{(2^2 - 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{(2^2 - 1)}$$

$$= \frac{(2^4 - 1)(2^4 + 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3}$$

$$= \frac{(2^8 - 1)(2^8 + 1)(2^{16} + 1) \dots (2^{128} + 1)}{3}$$

$$= \frac{(2^{256} - 1)}{3}$$

$$\text{Sol.10.(d)} \quad \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{110} < \frac{5}{6}$$

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots$$

$$\left(\frac{1}{10} - \frac{1}{11}\right) < \frac{5}{6} = 1 - \frac{1}{11} = \frac{10}{11} > \frac{5}{6}$$

Hence, the first statement is wrong.

$$\begin{aligned} \text{now, } \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{143} &> \frac{7}{13} \\ &= \frac{1}{2} \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) \right. \\ &\quad \left. + \dots + \left(\frac{1}{11} - \frac{1}{13}\right) \right] > \frac{7}{13} = \frac{1}{2} \left[\left(1 - \frac{1}{13}\right) \right] \\ &= \frac{6}{13} < \frac{7}{13} \end{aligned}$$

Hence, the second statement is also wrong.

Sol.11.(c)

$$\begin{aligned} &\left[\frac{5}{8} - \left\{ \frac{3}{8} - \left(\frac{5}{8} - \frac{3}{8} \right) \right\} \right] \text{ of } 8.8 - 1.2 \\ &= \frac{4\frac{1}{6} \div 2.5 \times 2 \div \frac{1}{6} \text{ of } 60 + \left(\frac{3}{4} - \frac{3}{8} \right)}{\left[\frac{5}{8} - \left\{ \frac{3}{8} - \frac{1}{4} \right\} \right] \text{ of } 8.8 - 1.2} \\ &= \frac{\frac{25}{6} \div 2.5 \times 2 \div 10 + \frac{3}{8}}{\left[\frac{5}{8} - \frac{1}{8} \right] \text{ of } 8.8 - 1.2} = \frac{\frac{1}{2} \text{ of } 8.8 - 1.2}{\frac{25}{6} \div \frac{5}{2} \times 2 \div 10 + \frac{3}{8}} = \frac{\frac{5}{3} \times \frac{1}{5} + \frac{3}{8}}{\frac{4.4 - 1.2}{\frac{1}{3} + \frac{3}{8}} = \frac{3.2}{\frac{17}{24}} = 4\frac{44}{85}} \end{aligned}$$

Sol.12.(c)

$$\begin{aligned} &\frac{24 \div \frac{3}{8} \text{ of } (8 + 2 \times \overline{7-3}) + \left[\frac{2}{11} \div \frac{4}{55} - \left\{ \frac{5}{8} + \frac{6}{16} \right\} \right]}{32 \div \overline{15-7} + 75 \div (6 + 15 \div 3 + 4)} \\ &= \frac{24 \div \frac{3}{8} \text{ of } (8 + 2 \times 4) + \left[\frac{5}{2} - 1 \right]}{32 \div 8 + 75 \div (6 + 5 + 4)} \\ &= \frac{24 \div \frac{3}{8} \text{ of } 16 + 1.5}{4 + 75 \div 15} = \frac{4 + 1.5}{9} \\ &= \frac{5.5}{9} = \frac{11}{18} \end{aligned}$$

Sol.13.(d)

$$\begin{aligned} &\frac{(6.9)^3 + (2.7)^3 + (5.8)^3 - 3 \times 6.9 \times 2.7 \times 5.8}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)} \\ &= \frac{(6.9+2.7+5.8)[(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9 \times 2.7) - (2.7 \times 5.8) - (5.8 \times 6.9)]}{(6.9)^2 + (2.7)^2 + (5.8)^2 - (6.9)(2.7) - (2.7)(5.8) - (5.8)(6.9)} \\ &= 6.9 + 2.7 + 5.8 = 15.4 \end{aligned}$$

Sol.14.(d) $9\frac{1}{5} \div \left\{ \left(16\frac{1}{5} \div 12\frac{2}{3} - 9\frac{2}{3} \right) \right.$

$$\begin{aligned} &\quad \left. + 17\frac{1}{2} \text{ of } 3\frac{1}{21} \right\} \\ &= \frac{46}{5} \div \left\{ \frac{81}{5} \div \left(\frac{38}{3} - \frac{29}{3} \right) + \frac{35}{2} \text{ of } \frac{64}{21} \right\} \\ &= \frac{46}{5} \div \left\{ \frac{81}{5} \div 3 + \frac{160}{3} \right\} \\ &= \frac{46}{5} \div \left\{ \frac{27}{5} + \frac{160}{3} \right\} \\ &= \frac{46}{5} \div \frac{881}{15} \Rightarrow \frac{46}{5} \times \frac{15}{881} \Rightarrow \frac{138}{881} \end{aligned}$$

Sol.15.(d)

$$\begin{aligned} &\frac{1}{4} + \frac{[(20.35)^2 - (8.35)^2] \times 0.0175}{(1.05)^2 + (1.05)(27.65)} \\ &= \frac{1}{4} + \frac{[(20.35+8.35)(20.35-8.35)] \times 0.0175}{(1.05)\{ (1.05) + (27.65) \}} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{4} + \frac{[28.7 \times 12] \times 0.0175}{(1.05)(28.7)} \\ &= \frac{1}{4} + \frac{12 \times 175}{105 \times 100} = \frac{1}{4} + \frac{12 \times 5}{3 \times 100} \\ &= \frac{1}{4} + \frac{1}{5} = \frac{9}{20} \end{aligned}$$

Sol.16.(a)

$$\begin{aligned} &\frac{119 - [48 \div 6 - 7 \{5 \times 12 \div 3 - (15 - 3 - 8)\}]}{24 \div 3 - 9 \text{ of } 3 + (52 - 8 - 4) \div 6} \\ &= \frac{119 - [8 - 7 \{5 \times 4 - (15 + 5)\}]}{8 - 27 + (52 - 4) \div 6} \\ &= \frac{119 - [8 - 7 \{20 - (20)\}]}{8 - 27 + (48) \div 6} \\ &= \frac{119 - [8 - 7 \{0\}]}{8 - 27 + 8} \\ &= \frac{119 - [8]}{-11} = -\frac{111}{11} \end{aligned}$$

Sol.17.(c) $\frac{1}{4 - \sqrt{8}}$ (on rationalizing)

$$\begin{aligned} &= \frac{1}{4 - \sqrt{8}} \times \frac{4 + \sqrt{8}}{4 + \sqrt{8}} = \frac{4 + \sqrt{8}}{8} \text{ so,} \\ &= \frac{4 + \sqrt{8}}{8} + \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} - \frac{3 - 2\sqrt{2}}{3 + 2\sqrt{2}} \\ &= \frac{4 + \sqrt{8}}{8} + \frac{(3 + 2\sqrt{2})^2 - (3 - 2\sqrt{2})^2}{(3)^2 - (2\sqrt{2})^2} \\ &= \frac{4 + \sqrt{8}}{8} + \frac{24\sqrt{2}}{9 - 8} \Rightarrow \frac{1}{2} + \frac{\sqrt{2}}{4} + 24\sqrt{2} \end{aligned}$$

Since, $\frac{1}{2} + \frac{97}{4}\sqrt{2} = a + b\sqrt{2}$

On comparing $a = \frac{1}{2}$, $b = \frac{97}{4}$

$$3a + 4b = 3 \times \frac{1}{2} + 4 \times \frac{97}{4} = 98\frac{1}{2}$$

Sol.18.(a) After checking option one by one option (a) satisfies.

$$\begin{aligned} &4 \times 30 - 15 \div 5 + 12 = 21 \\ &\Rightarrow 4 \times 30 \div 5 - 15 + 12 = 21 \\ &\Rightarrow 4 \times 6 - 15 + 12 = 21 \\ &\Rightarrow 24 - 15 + 12 = 21 \Rightarrow 9 + 12 = 21 \\ &\Rightarrow 21 = 21 \end{aligned}$$

Sol.19.(c) $\frac{4}{1 + \sqrt{2} + \sqrt{3}}$

$$\begin{aligned} &= a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6} \\ &= \frac{4}{1 + \sqrt{2} + \sqrt{3}} \times \frac{\sqrt{3} + 1 - \sqrt{2}}{\sqrt{3} + 1 - \sqrt{2}} \\ &= (a + b)(a - b) = a^2 - b^2 \\ &= \frac{4\sqrt{3} + 4 - 4\sqrt{2}}{(\sqrt{3} + 1)^2 - (\sqrt{2})^2} = \frac{4\sqrt{3} + 4 - 4\sqrt{2}}{2\sqrt{3} + 2} \\ &= \frac{2\sqrt{3} + 2 - 2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1} \\ &= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{3 - 1} \\ &= \frac{(2\sqrt{3} + 2 - 2\sqrt{2})(\sqrt{3} - 1)}{2} \\ &= (\sqrt{3} + 1 - \sqrt{2})(\sqrt{3} - 1) \\ &= 3 + \sqrt{3} - \sqrt{6} - \sqrt{3} - 1 + \sqrt{2} \\ &= 2 + \sqrt{2} - \sqrt{6} = a + b\sqrt{2} + c\sqrt{3} - d\sqrt{6} \\ &a = 2; b = 1; c = 0; d = 1 \end{aligned}$$

$$a + b + c + d = 2 + 1 + 0 + 1 = 4$$

Sol.20.(a)

$$\begin{aligned} &\left[\left(\frac{2}{3} \right)^3 (2x + 3) \right]^{\frac{-3}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} (3x + 7) \right]^{\frac{-6}{5}} \\ &\Rightarrow \left[\left(\frac{2}{3} \right)^3 (2x + 3) \right]^{\frac{3}{4}} = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} (3x + 7) \right]^{\frac{6}{5}} \\ &\Rightarrow \left[\left(\frac{2}{3} \right)^3 \frac{3(2x + 3)}{4} \right] = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} \frac{6(3x + 7)}{5} \right] \\ &\Rightarrow \left[\left(\frac{2}{3} \right)^3 \frac{6x + 9}{4} \right] = \left[\left(\frac{2}{3} \right)^{\frac{2}{3}} \frac{18x + 42}{5} \right] \\ &\Rightarrow \left(\frac{2}{3} \right)^{\frac{3(6x + 9)}{4}} = \left(\frac{2}{3} \right)^{\frac{2(18x + 42)}{3 \times 5}} = \left(\frac{2}{3} \right)^{\frac{18x + 27}{4}} \\ &= \frac{36x + 84}{15} = \frac{18x + 27}{4} \\ &\Rightarrow 4(36x + 84) = 15(18x + 27) \\ &\Rightarrow 144x + 336 = 270x + 405 \\ &\Rightarrow 126x = 336 - 405 \Rightarrow 126x = -69 \\ &\Rightarrow x = -\frac{69}{126} = -\frac{23}{42} \end{aligned}$$

Now put the value of x in given equation

$$\begin{aligned} \sqrt{2 - 42x} &= \sqrt{2 - 42\left(-\frac{23}{42}\right)} \\ &= \sqrt{2 + 23} = \sqrt{25} = 5 \end{aligned}$$

Sol.21.(b) $\frac{42 - 12 \times 3 + 8 \div 2 + 15}{8 \times 2 - 4 + 9 \div 3}$

After applying changes as per question we get

$$\frac{42 + 12 \div 3 - 8 \times 2 - 15}{8 \div 2 + 4 - 9 \times 3} = -\frac{15}{19}$$

Sol.22.(d) $\frac{(4.6)^4 + (5.4)^4 + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84}$

$$\begin{aligned} &= \frac{\{(4.6)^2 + (5.4)^2\}^2 - 2\{(4.6)^2(5.4)^2\} + (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84} \\ &= \frac{\{(4.6)^2 + (5.4)^2\}^2 - (24.84)^2}{(4.6)^2 + (5.4)^2 + 24.84} \\ &= \frac{[(4.6)^2 + (5.4)^2 - (24.84)] \{[(4.6)^2 + (5.4)^2 + (24.84)]\}}{(4.6)^2 + (5.4)^2 + 24.84} \\ &= (4.6)^2 + (5.4)^2 - 24.84 \\ &= (4.6 + 5.4)^2 - 2(4.6)(5.4) - 24.84 \\ &= 100 - 74.52 = 25.48 \end{aligned}$$

Sol.23.(a) $\frac{(0.545)(0.081)(0.51)(5.2)}{(0.324)^3 + (0.221)^3 - (0.545)^3}$

$$\begin{aligned} &= \frac{(0.545)(0.081) \times 4 \times 3 \times (0.17)(1.3)}{(0.324)^3 + (0.221)^3 - (0.545)^3} \\ &= \frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3} \end{aligned}$$

Now, since

$$\begin{aligned} &(0.324 + 0.221 - 0.545 = 0) \\ &\Rightarrow (0.324)^3 + (0.221)^3 - (0.545)^3 = \\ &\quad - \{3 \times (0.545)(0.324)(0.221)\} \\ &\quad \dots (\text{Algebraic property}) \end{aligned}$$

$$\begin{aligned} \text{So, } &\frac{3 \times (0.545)(0.324)(0.221)}{(0.324)^3 + (0.221)^3 - (0.545)^3} \\ &= \frac{3 \times (0.545)(0.324)(0.221)}{-\{3 \times (0.545)(0.324)(0.221)\}} = -1 \end{aligned}$$

$$\begin{aligned}\text{Sol.24.(d)} \quad & 22.\overline{4} + 11.\overline{567} - 33.\overline{59} \\ & = 22.444444\dots\infty + 11.5676767\dots\infty - \\ & \quad 33.599999\dots\infty \\ & = 0.412121212\dots\infty = 0.\overline{412}\end{aligned}$$

$$\begin{aligned}\text{Sol.25.(d)} \quad & \frac{(0.7)^2 \div 0.14 + (0.6)^2 \div 0.18 + (0.5)^2 \div 0.05}{4(2.5 \text{ of } 4 - 13 \times 0.25 \times 3)} \\ & = \frac{0.49 \div 0.14 + 0.36 \div 0.18 + 0.25 \div 0.05}{4(10 - 9.75)} \\ & = \frac{3.5 + 2 + 5}{4 \times 0.25} = 10.5 = \frac{21}{2}\end{aligned}$$

$$\begin{aligned}\text{Sol.26.(d)} \quad & x = \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} + \dots + \frac{1}{23.24} \\ & = \frac{1}{12} - \frac{1}{13} + \frac{1}{13} - \frac{1}{14} + \dots + \frac{1}{23} - \frac{1}{24} \\ & = \frac{1}{12} - \frac{1}{24} = \frac{1}{24} \\ & y = \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} + \dots + \frac{1}{71.72} \\ & = \frac{1}{36} - \frac{1}{37} + \frac{1}{37} - \frac{1}{38} + \dots + \frac{1}{71} - \frac{1}{72} \\ & = \frac{1}{36} - \frac{1}{72} = \frac{1}{72} \\ \text{Therefore, } \frac{x}{y} &= \frac{\frac{1}{24}}{\frac{1}{72}} = 3\end{aligned}$$

Alternate Method :-

$$\begin{aligned}x &= \frac{1}{12.13} + \frac{1}{13.14} + \frac{1}{14.15} + \dots + \frac{1}{23.24} \\ &= \frac{1}{13-12} \left(\frac{1}{12} - \frac{1}{24} \right) = \frac{1}{24} \\ y &= \frac{1}{36.37} + \frac{1}{37.38} + \frac{1}{38.39} + \dots + \frac{1}{71.72} \\ &= \frac{1}{37-36} \left(\frac{1}{36} - \frac{1}{72} \right) = \frac{1}{72} \\ \text{Therefore, } \frac{x}{y} &= \frac{\frac{1}{24}}{\frac{1}{72}} = 3\end{aligned}$$

$$\begin{aligned}\text{Sol.27.(a)} \quad & \frac{0.01404}{24^2 + 6^2 - 144} \\ & = \frac{0.01404}{576 + 36 - 144} = \frac{0.01404}{612 - 144} \\ & = \frac{0.01404}{468} = 3 \times 10^{-5}\end{aligned}$$

$$\begin{aligned}\text{Sol.28.(b)} \quad & \frac{1.0025 + 6.25 \times 10^{-6}}{0.0025 + 0.95} \\ & = \left(\frac{1.00250625}{0.9525} \right) = 1.0525\end{aligned}$$

$$\begin{aligned}\text{Sol.29.(d)} \quad & \frac{9}{15} \text{ of } \left(\frac{2}{3} \div \frac{2}{3} \text{ of } \frac{3}{2} \right) \div \left(\frac{3}{4} \times \frac{3}{4} \div \frac{3}{4} \right. \\ & \left. \text{of } \frac{4}{3} \right) \text{ of } \left(\frac{5}{4} \div \frac{5}{2} \times \frac{2}{5} \text{ of } \frac{4}{5} \right) \\ & = \frac{9}{15} \text{ of } \frac{2}{3} \div \frac{9}{16} \text{ of } \frac{4}{25} \\ & = \frac{2}{5} \div \frac{9}{100} = \frac{2}{5} \times \frac{100}{9} = \frac{40}{9}\end{aligned}$$

$$\begin{aligned}\text{Sol.30.(c)} \quad & (5 + 3 \div 5 \times 5) / (3 \div 3 \text{ of } 6) \text{ of } \\ & (4 \times 4 \div 4 \text{ of } 4 + 4 \div 4 \times 4) \\ & = \frac{(5 + 3 \div 5 \times 5)}{(3 \div 3 \text{ of } 6) \text{ of } (4 \times 4 \div 4 \text{ of } 4 + 4 \div 4 \times 4)} \\ & = \frac{8}{\frac{1}{6} \text{ of } 5} = \frac{48}{5} = 9\frac{3}{5}\end{aligned}$$

$$\begin{aligned}\text{Sol.31.(d)} \quad & 2\frac{7}{8} \div (3\frac{5}{6} \div \frac{2}{7} \text{ of } 2\frac{1}{3}) \times \\ & [(2\frac{6}{7} \text{ of } 4\frac{1}{5} \div \frac{2}{3}) \times \frac{5}{9}] \\ & = \frac{23}{8} \div (\frac{23}{6} \div \frac{2}{3}) \times [10] = 5\end{aligned}$$

$$\begin{aligned}\text{Sol.32.(d)} \quad & 2 \times 3 \div 2 \text{ of } 3 \times 2 \div (4 + 4 \\ & \times 4 \div 4 \text{ of } 4 - 4 \div 4 \times 4) \\ & = 2 \times 3 \div 6 \times 2 \div (4 + 4 \times 4 \div 16 - 4 \\ & \quad \div 4 \times 4) \\ & = 2 \times \frac{1}{2} \times 2 \times 1 = 2\end{aligned}$$

$$\begin{aligned}\text{Sol.33.(d)} \quad & \text{Given fraction is} \\ & \frac{5.75 \times 5.75 \times 5.75 + 3.25 \times 3.25 \times 3.25}{57.5 \times 57.5 + 32.5 \times 32.5 - 57.5 \times 32.5} \\ & = \frac{5.75 \times 5.75 \times 5.75 + 3.25 \times 3.25 \times 3.25}{(5.75 \times 5.75 + 3.25 \times 3.25 - 5.75 \times 3.25) \times 100} \\ & = \frac{(5.75 + 3.25)(5.75 \times 5.75 + 3.25 \times 3.25 - 5.75 \times 3.25)}{(5.75 \times 5.75 + 3.25 \times 3.25 - 5.75 \times 3.25) \times 100} \\ & = \frac{9}{100} = 0.09\end{aligned}$$

$$\text{Sol.34.(c)} \quad X = \left(\frac{3}{2}\right)^2 \times \left(\frac{2}{3}\right)^{-4}$$

$$X = \left(\frac{3}{2}\right)^2 \times \left(\frac{3}{2}\right)^4 = \left(\frac{3}{2}\right)^6$$

$$X^{-1} = \left(\frac{2}{3}\right)^6 \Rightarrow X^{-2} = \left(\frac{2}{3}\right)^{12}$$

$$\begin{aligned}\text{Sol.35.(c)} \quad & \sqrt[3]{0.08 \times 0.8 \times p} = 0.008 \times 0.8 \times \sqrt[3]{q} \\ & 8 \times 8 \times 10^{-3} \times p = (8 \times 8 \times 10^{-4})^3 \times q \\ & 8^2 \times 10^{-3} \times p = 8^6 \times 10^{-12} \times q \\ & \frac{p}{q} = 8^4 \times 10^{-9}\end{aligned}$$

$$\begin{aligned}\text{Sol.36.(b)} \quad & \left(\sqrt[3]{343}\right)^3 \times \left(\sqrt[3]{216}\right)^3 \\ & = \sqrt[3]{343} \times \sqrt[3]{216} = 7 \times 6 = 42\end{aligned}$$

$$\begin{aligned}\text{Sol.37.(b)} \quad & \sqrt{625} + \sqrt{\frac{625}{100}} + \sqrt{\frac{625}{10000}} + \sqrt{\frac{625}{1000000}} \\ & = 25 + \frac{25}{10} + \frac{25}{100} + \frac{25}{1000} \\ & = 25 + 2.5 + 0.25 + 0.025 \\ & = 27.775\end{aligned}$$

$$\begin{aligned}\text{Sol.38.(b)} \quad & \left[\frac{1}{8} + \left\{ \frac{1}{6} \times \left(\frac{36}{45} \div \frac{24}{25} \right) - \left(\frac{12}{21} \times \frac{14}{15} \div \right. \right. \right. \\ & \left. \left. \frac{24}{45} \right\} + \frac{27}{36} \right] \\ & = \left[\frac{1}{8} + \left\{ \frac{1}{6} \times \frac{5}{6} - \left(\frac{12}{21} \times \frac{7}{4} \right) \right\} + \frac{27}{36} \right] \\ & = \left[\frac{1}{8} + \left\{ \frac{5}{36} - 1 \right\} + \frac{27}{36} \right]\end{aligned}$$

$$= \left[\frac{1}{8} - \frac{31}{36} + \frac{27}{36} \right] = \frac{1}{72}$$

$$\begin{aligned}\text{Sol.39.(a)} \quad & \frac{\sqrt{1.24} \times \sqrt{2.79}}{\sqrt{2.64} \times \sqrt{5.94}} = \sqrt{\frac{1.24 \times 2.79}{2.64 \times 5.94}} \\ & = \sqrt{\frac{31 \times 31}{66 \times 66}} = \frac{31}{66}\end{aligned}$$

$$\begin{aligned}\text{Sol.40.(c)} \quad & = \frac{0.04}{0.05} \text{ of } \frac{\left(3\frac{1}{3} - 2\frac{1}{2}\right) \div \frac{1}{2} \text{ of } 1\frac{1}{4}}{\frac{1}{3} + \frac{1}{5} \text{ of } \frac{1}{9}} \\ & = \frac{4}{5} \text{ of } \frac{\left(\frac{10}{3} - \frac{5}{2}\right) \div \frac{1}{2} \text{ of } \frac{5}{4}}{\frac{1}{3} + \frac{1}{5} \text{ of } \frac{1}{9}} \\ & = \frac{4}{5} \text{ of } \frac{\left(\frac{5}{6}\right) \div \frac{5}{8}}{\frac{1}{3} + \frac{1}{45}} \\ & = \frac{4}{5} \text{ of } \frac{\left(\frac{4}{3}\right)}{\frac{16}{45}} = \frac{4}{5} \times \frac{15}{4} = 3\end{aligned}$$

$$\begin{aligned}\text{Sol.41.(d)} \quad & (48 \div 72 \times 3) - [15 \div 8 \times (40 - \\ & 32) - 10] + 2P = 6 \div 2 \\ \Rightarrow (48 \div 72 \times 3) - [15 \div 8 \times 8 - 10] + 2P &= 6 \\ \div 2 \\ \Rightarrow (48 \div 72 \times 3) - \left[\frac{15}{8} \times 8 - 10\right] + 2P &= 6 \div 2 \\ \Rightarrow \left(\frac{2}{3} \times 3\right) - [15 - 10] + 2P &= 6 \div 2 \\ \Rightarrow 2 - 5 + 2P &= 3 \Rightarrow P = 3\end{aligned}$$

$$\begin{aligned}\text{Sol.42.(c)} \quad & \frac{1 - \frac{3}{7} \div \frac{9}{49} \times \frac{3}{14}}{1 + \frac{3}{5} \div \frac{1}{25} \times \frac{3}{30}} \\ & = \frac{1 - \frac{3}{7} \times \frac{49}{9} \times \frac{3}{14}}{1 + \frac{3}{5} \times \frac{25}{1} \times \frac{1}{30}} = \frac{1 - \frac{1}{2}}{1 + \frac{1}{2}} = \frac{1}{3}\end{aligned}$$

$$\begin{aligned}\text{Sol.43.(a)} \quad & 1\frac{2}{5} - [3\frac{3}{4} \div \{1\frac{1}{4} \div \frac{1}{2}(1\frac{1}{2} \times 3\frac{1}{3} \div 1\frac{1}{3})\}] \\ & = \frac{7}{5} - \left[\frac{15}{4} \div \left\{ \frac{5}{4} \div \frac{1}{2} \left(\frac{3}{2} \times \frac{10}{3} \div \frac{4}{3} \right) \right\} \right] \\ & = \frac{7}{5} - \left[\frac{15}{4} \div \left\{ \frac{5}{4} \div \frac{1}{2} \left(\frac{3}{2} \times \frac{10}{3} \times \frac{3}{4} \right) \right\} \right] \\ & = \frac{7}{5} - \left[\frac{15}{4} \div \left\{ \frac{5}{4} \div \frac{1}{2} \times \frac{15}{4} \right\} \right] \\ & = \frac{7}{5} - \left[\frac{15}{4} \div \left\{ \frac{5}{4} \times \frac{2}{1} \times \frac{15}{4} \right\} \right] \\ & = \frac{7}{5} - \left[\frac{15}{4} \div \left\{ \frac{75}{8} \right\} \right] = \frac{7}{5} - \left[\frac{15}{4} \times \frac{8}{75} \right] \\ & = \frac{7}{5} - \left[\frac{2}{5} \right] = \frac{7-2}{5} = \frac{5}{5} = 1\end{aligned}$$

$$\begin{aligned}\text{Sol.44.(c)} \quad & \frac{1\frac{1}{2} \div 3\frac{1}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}}{\frac{1}{5} \times 3\frac{1}{2} - \frac{1}{3} \div 1\frac{3}{4} \times 3\frac{1}{2}} \\ & = \frac{\frac{3}{2} \div \frac{13}{4} + \frac{1}{2} \div \frac{13}{14} + \frac{1}{5}}{\frac{1}{5} \times \frac{7}{2} - \frac{1}{3} \div \frac{7}{4} \times \frac{7}{2}}\end{aligned}$$

$$= \frac{\frac{3}{2} \times \frac{4}{13} + \frac{1}{2} \times \frac{14}{13} + \frac{1}{5}}{\frac{1}{5} \times \frac{7}{2} - \frac{1}{3} \times \frac{4}{7} \times \frac{7}{2}}$$

$$= \frac{\frac{6}{13} + \frac{7}{13} + \frac{1}{5}}{\frac{7}{10} - \frac{2}{3}} = \frac{\frac{6}{5}}{\frac{1}{30}} = 36$$

Sol.45.(b)

$$\Rightarrow b + 2a - [(3b + a) - (2a + b) + 2a] - a$$

$$\Rightarrow b + 2a - [3b + a - 2a - b + 2a] - a$$

$$\Rightarrow b + 2a - [2b + a] - a$$

$$\Rightarrow b + 2a - 2b - a - a = -b$$

Sol.46.(b)

$$24 \div 6 \times \left(\frac{3}{4} \text{ of } \frac{1}{3} + \frac{5}{3} \text{ of } \frac{3}{4}\right) = ?$$

$$4 \times \left(\frac{1}{4} + \frac{5}{4}\right) \Rightarrow 4 \times \frac{6}{4} = 6$$

Sol.47.(d) $71 - (-3)\{-2 - (\overline{8 - 3})\} + 3\{5 + (-2)(-1)\}$
 $\Rightarrow 71 - (-3)\{-2 - 5\} + 3\{5 + 2\}$
 $\Rightarrow 71 - (-3)\{-7\} + 3\{7\}$
 $\Rightarrow 71 - 21 + 21 = 71$

Sol.48.(b) $15 \div 5 \text{ of } 3 \times 4 + 8 \div 12 \text{ of } 4 \times 12 - 4 \div 5 \times 20$
 $= 15 \div 15 \times 4 + 8 \div 48 \times 12 - 4 \div 5 \times 20$
 $= 1 \times 4 + \frac{1}{6} \times 12 - \frac{4}{5} \times 20$
 $= 4 + 2 - 16 = -10$

Sol.49.(a) $\frac{3 \div 3 \times 2 - 6 \div 3 + 7}{5 \div 5 \times 6 \div 6 + 3 + 3}$
 $\frac{2 - 2 + 7}{1 + 3 + 3} \Rightarrow \frac{7}{7} = 1$

Sol.50.(c) $3\frac{1}{2} - [2\frac{1}{4} \div \{1\frac{1}{4} - \frac{3}{4}(1\frac{1}{2} - \frac{1}{2} - 2\frac{1}{2})\}]$
 $= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} - \frac{3}{4}(\frac{3}{2} - \frac{1}{2} - \frac{5}{2})\}]$
 $= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} - \frac{3}{4}(-\frac{3}{2})\}]$
 $= \frac{7}{2} - [\frac{9}{4} \div \{\frac{5}{4} + \frac{9}{8}\}]$
 $= \frac{7}{2} - \frac{18}{19} = \frac{97}{38} = 2\frac{21}{38}$

Sol.51.(d)

$$10(8 + 4R) \div (12 \times 5) - 5 \times 2^3 = 6$$

$$\Rightarrow \frac{(4 + 2R)}{3} - 40 = 6$$

$$\Rightarrow \frac{(4 + 2R)}{3} = 46$$

$$\Rightarrow (4 + 2R) = 138$$

$$\Rightarrow R = \frac{138 - 4}{2} = 67$$

Sol.52.(b)

$$[1^2 + 2^2 + 3^2 + 4^2] \times \frac{2}{3 \times 5} \div \left(\frac{1}{4}\right)^2$$

$$\Rightarrow [1 + 4 + 9 + 16] \times \frac{2}{15} \times 16$$

$$\Rightarrow 30 \times \frac{2}{15} \times 16 = 64$$

Sol.53.(d)

$$\left[\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5}\right] \times \frac{1}{5} + \left(1 - \frac{6}{25}\right)$$

$$\left[\frac{1}{9} + \frac{1}{9} \times \frac{1}{9}\right] \times \frac{1}{10}$$

$$= \left[\frac{1 \times \frac{1}{5}}{\frac{1}{9} + \frac{1}{81}}\right] \times \frac{1}{10} + \left(\frac{19}{25}\right)$$

$$= \frac{\frac{1}{25}}{\left[\frac{10}{81}\right] \times \frac{1}{10}} + \left(\frac{19}{25}\right)$$

$$= \frac{1}{25} \times \frac{81}{1} + \left(\frac{19}{25}\right)$$

$$= \frac{81 + 19}{25} = \frac{100}{25} = 4$$

Sol.54.(a) $\left(\frac{5}{3} + \frac{6}{5} \div \frac{18}{5}\right) \text{ of } \frac{5}{12} - \left(\frac{5}{6} \div \frac{5}{12}\right) \text{ of } \frac{5}{12} + \frac{2}{3} \text{ of } \frac{3}{2}$
 $\Rightarrow \left(\frac{5}{3} + \frac{6}{5} \times \frac{5}{18}\right) \text{ of } \frac{5}{12} - \left(\frac{5}{6} \times \frac{12}{5}\right)$
 $\text{of } \frac{5}{12} + \frac{2}{3} \times \frac{3}{2}$
 $\Rightarrow \left(\frac{5}{3} + \frac{1}{3}\right) \text{ of } \frac{5}{12} - (2) \text{ of } \frac{5}{12} + 1$
 $\Rightarrow (2) \text{ of } \frac{5}{12} - \frac{5}{6} + 1 = \frac{5}{6} - \frac{5}{6} + 1 = 1$

Sol.55.(c) $\{[2^4(20 \times 5 + 10) \div 11] - (2^3 \times 15 \div 20) - (9 \times 16)\}$
 $\Rightarrow \{[2^4(110) \div 11] - (6) - (144)\}$
 $\Rightarrow [160 - 6 - 144] \Rightarrow 160 - 150 = 10$

Sol.56.(a) $6\frac{4}{5} \div [4\frac{1}{4} \div \{5\frac{1}{3} \times \frac{3}{8} + (3\frac{1}{2} + \frac{2}{3}) \times \frac{4}{5}\}]$
 $= \frac{34}{5} \div [\frac{17}{4} \div \{\frac{16}{3} \times \frac{3}{8} + (\frac{7}{2} + \frac{2}{3}) \times \frac{4}{5}\}]$
 $= \frac{34}{5} \div [\frac{17}{4} \div \{2 + (\frac{25}{6}) \times \frac{4}{5}\}]$
 $= \frac{34}{5} \div [\frac{17}{4} \div \{2 + (\frac{10}{3})\}]$
 $= \frac{34}{5} \div [\frac{17}{4} \div \frac{16}{3}]$
 $= \frac{34}{5} \times \frac{64}{51} = \frac{128}{15} = 8\frac{8}{15}$

Sol.57.(a)

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\frac{4[(17)^3 - (7)^3]}{(17)^2 + (7)^2 + 119} = 4(17 - 7) = 40$$

Sol.58.(c) $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$

$$\frac{(612 + 388)^2 + (612 - 388)^2}{(612 \times 612) + (388 \times 388)}$$

$$= \frac{2 \times \{(612)^2 + (388)^2\}}{(612 \times 612) + (388 \times 388)} = 2$$

Sol.59.(b) $87 \times 256 + 16 - 77 \div 1359$

After changing sign ,
 $87 + 256 \div 16 \times 77 - 1359$
 $= 87 + 16 \times 77 - 1359$
 $= 87 + 1232 - 1359 = -40$

Sol.60.(d) $\frac{a^3 + b^3}{a^2 - ab + b^2} = (a + b)$
 $\frac{(0.96)^3 + (0.04)^3}{(0.96)^2 - (0.96 \times 0.04) + (0.04)^2}$
 $= (0.96 + 0.04) = 1$

Sol.61.(d) $\sqrt{a} = 3b \Rightarrow b^2 = \frac{a}{9}$

Now, $\frac{a}{b^2} = \frac{9}{1} = 9$

Sol.62.(c)

$$\frac{(0.25)^4 + 2 \times (0.25)^2 + 1 - (0.25)^2}{(0.25)^2 + 0.25 + 1}$$

$$= \frac{\{(0.25)^2 + 1\}^2 - (0.25)^2}{(0.25)^2 + 0.25 + 1}$$

$$= \frac{\{(0.25)^2 + 1 + 0.25\}\{(0.25)^2 + 1 - 0.25\}}{(0.25)^2 + 0.25 + 1}$$

$$= \{(0.25)^2 + 1 - 0.25\}$$

$$= \{0.0625 + 1 - 0.25\} = 0.8125$$

Sol.63.(b) $\frac{3 + 8 \times 8 \div 8 \text{ of } 8 + 7 \div 7 \times 2}{5 \div 5 \text{ of } 4 + 3 \times 3 \div 3 - 3 + 2}$
 $= \frac{3 + 8 \times 8 \div 64 + 7 \div 7 \times 2}{5 \div 20 + 3 \times 3 \div 3 - 3 + 2}$
 $= \frac{3 + 1 + 1 \times 2}{\frac{1}{4} + 3 - 3 + 2} = \frac{6}{\frac{1}{4} + 2}$
 $= \frac{24}{9} = 2\frac{2}{3}$

Sol.64.(b)

$$\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$$

$$= \frac{5 + 2\sqrt{6}}{3 - 2} = 5 + 2 \times 2.45 = 9.9$$

Sol.65.(b) $[\frac{1}{5} \text{ of } (35 - 5) + 4]^2 -$

$$[9 - \{7 + (2 - 3)\} + 5]^2$$

$$= [6 + 4]^2 - [9 - \{7 - 1\} + 5]^2$$

$$= 100 - [9 - 6 + 5]^2 = 100 - 64 = 36$$

Sol.66.(b)

$$\frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)$$

So,

$$\frac{(80)^3 + (70)^3 + (50)^3 - 3 \times 80 \times 70 \times 50}{(80)^2 + (70)^2 + (50)^2 - 80 \times 70 - 70 \times 50 - 50 \times 80}$$

$$= (80 + 70 + 50) = 200$$

Sol.67.(d)

$$\frac{\{4\frac{1}{3} + 3\frac{1}{3} \times 1\frac{4}{5} \div 3\frac{3}{4} \times (1\frac{1}{2} + 1\frac{1}{3})\}}{(\frac{2}{3} \div \frac{5}{6} \times \frac{2}{3})}$$

$$\frac{\{\frac{13}{3} + \frac{10}{3} \times \frac{9}{5} \div \frac{15}{4} \times (\frac{3}{2} + \frac{4}{3})\}}{(\frac{4}{5} \times \frac{2}{3})}$$

$$\frac{\{\frac{13}{3} + \frac{10}{3} \times \frac{9}{5} \div \frac{15}{4} \times \frac{17}{6}\}}{(\frac{8}{15})}$$

$$= \left\{ \frac{13}{3} + \frac{68}{15} \right\} \times \frac{15}{8} = \frac{133}{15} \times \frac{15}{8}$$

$$= \frac{133}{8} = 16\frac{5}{8}$$

Sol.68.(a)

$$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$\frac{(379 + 276)^2 + (379 - 276)^2}{379 \times 379 + 276 \times 276}$$

$$= \frac{2(379^2 + 276^2)}{379^2 + 276^2} = 2$$

Sol.69.(d) $25(4 + 3p) \div 5 - 3 \times 8 = 26$
 $\Rightarrow 5(4 + 3p) - 24 = 26 \Rightarrow 5(4 + 3p) = 50$
 $\Rightarrow 3p = 10 - 4 \Rightarrow p = 2$

Sol.70.(d) $\frac{(381 + 119)^2 + (381 - 119)^2}{(381)^2 + (119)^2}$

$$\frac{(381 + 119)^2 + (381 - 119)^2}{(381)^2 + (119)^2}$$

$$= \frac{2(381^2 + 119^2)}{381^2 + 119^2} = 2$$

Sol.71.(a) Formula used :-

$$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$$

$$\frac{(4.2)^3 - 0.008}{(4.2)^2 + 0.84 + 0.04}$$

$$= \frac{(4.2)^3 - (0.2)^3}{(4.2)^2 + 0.84 + 0.04}$$

$$= \frac{(4.2 - 0.2)[(4.2)^2 + 4.2 \times 0.2 + (0.2)^2]}{(4.2)^2 + 0.84 + 0.04}$$

$$= \frac{4[(4.2)^2 + 0.84 + 0.04]}{(4.2)^2 + 0.84 + 0.04} = 4$$

Sol.72.(b) $\frac{2 + 7 \times 7 \div 9 \text{ of } 9 + 6 \div 6 \times 2}{4 \div 4 \text{ of } 5 + 7 \times 7 \div 7 - 6 + 4}$

$$= \frac{2 + 49 \div 81 + 1 \times 2}{4 \div 20 + 7 - 6 + 4} = \frac{2 + \frac{49}{81} + 2}{\frac{1}{5} + 5}$$

$$= \frac{373}{81} \times \frac{5}{26} = \frac{1865}{2106}$$

Sol.73.(c) Formula :- $a^3 - b^3 = (a - b)$

$$(a^2 + ab + b^2)$$

$$\frac{143 \times 143 + 143 \times 139 + 139 \times 139}{143 \times 143 \times 143 - 139 \times 139 \times 139}$$

$$= \frac{(143)^2 + 143 \times 139 + (139)^2}{(143)^3 - (139)^3}$$

$$= \frac{(143)^2 + 143 \times 139 + (139)^2}{(143 - 139)[(143)^2 + 143 \times 139 + (139)^2]}$$

$$= \frac{1}{4}$$

Sol.74.(a) $\frac{10x^2 + 5x + 2xy + y}{5x + y}$

$$\frac{2x(5x + y) + (5x + y)}{5x + y} = \frac{(5x + y)(2x + 1)}{5x + y}$$

$$2x + 1 = 2 \times (-1) + 1 = -1$$

Sol.75.(d) $\left[\frac{85}{34} \times \frac{1}{18} - \left\{ \left(\frac{46}{69} \div \frac{27}{135} \right) - \right. \right.$

$$\left. \left(\frac{86}{129} \times \frac{91}{14} \right) \right\} \text{ of } \frac{112}{36}]$$

$$= \left[\frac{5}{2} \times \frac{1}{18} - \left\{ \left(\frac{2}{3} \times \frac{135}{27} \right) - \left(\frac{2}{3} \times \frac{13}{2} \right) \right\} \right.$$

$$\left. \text{ of } \frac{28}{9} \right]$$

$$= \left[\frac{5}{36} - \left\{ \left(\frac{2}{3} \times 5 \right) - \left(\frac{13}{3} \right) \right\} \text{ of } \frac{28}{9} \right]$$

$$= \frac{5}{36} - \{-1\} \text{ of } \frac{28}{9}$$

$$= \left[\frac{5}{36} + \frac{28}{9} \right] = \left[\frac{5 + 112}{36} \right] = \frac{13}{4} = 3\frac{1}{4}$$

Sol.76.(a) $18 \div 3 \text{ of } 2 \times 5 + 72 \div 18 \text{ of } 2 \times$
 $3 - 4 \div 8 \times 2$

$$= 18 \div 6 \times 5 + 72 \div 36 \times 3 - \frac{1}{2} \times 2$$

$$= 3 \times 5 + 2 \times 3 - 1 = 15 + 6 - 1 = 20$$

Sol.77.(d)

$$y + 2x - [y - (y - x + y) - (x + y) + y] - 2y$$

$$= y + 2x - [(x - y) - x] - 2y$$

$$= y + 2x - [-y] - 2y$$

$$= y + 2x + y - 2y = 2x$$

Sol.78.(b) $(a + b)^2 - (a - b)^2 = 4ab$

$$\frac{(326 + 222)^2 - (326 - 222)^2}{(326 \times 222)}$$

$$= \frac{(4 \times 326 \times 222)}{(326 \times 222)} = 4$$

Sol.79.(a) $[\sqrt{25} + 12 \div 3 - \{20 + (16 \text{ of } 8 \div$
 $16) - (54 \div 18 \text{ of } \frac{1}{2})\}]$

$$[5 + 4 - \{20 + (8) - (54 \div 9)\}]$$

$$[5 + 4 - \{20 + (8) - (6)\}]$$

$$5 + 4 - 22 = -13$$

Sol.80.(c) $\frac{3a + b}{2} - \frac{a - 3b}{3} + 2b$

$$= \frac{3(3a + b) - 2(a - 3b) + 12b}{6}$$

$$= \frac{9a + 3b - 2a + 6b + 12b}{6}$$

$$= \frac{7a + 21b}{6} = \frac{7(a + 3b)}{6}$$

Sol.81.(c) $(a - 2b)(b - 3a) + (a + b)(a - 3b) - (b - 3a)(4a - 5b)$

$$= -3a^2 - 2b^2 + 7ab + a^2 - 3b^2 - 2ab - (-12a^2 - 5b^2 + 19ab)$$

$$= (-3a^2 + a^2 + 12a^2) + (-2b^2 - 3b^2 + 5b^2) + (7ab - 2ab - 19ab)$$

$$= 10a^2 - 14ab$$

Sol.82.(b) $(3^2)^2 + 3 \times 3 \div 3 - 3$

$$= 81 + 3 \times 1 - 3 = 81 + 3 - 3 = 81 = 9^2$$

Sol.83.(a) $720 \div 8 + 915 \div 15 - m + 32 \times 5$

$$= 1104 \div 16 \times 111 \div 37$$

$$\Rightarrow 90 + 61 - m + 160 = 69 \times 3$$

$$\Rightarrow 311 - m = 207 \Rightarrow m = 104$$

Sol.84.(b) $(4x + 1)^2 - (4x + 3)(4x - 1)$

$$= 16x^2 + 1 + 8x - (16x^2 - 4x + 12x - 3)$$

$$= 16x^2 + 1 + 8x - 16x^2 + 4x - 12x + 3$$

$$= 4$$

Short Tricks :- Put $x = 0$,

$$(4x + 1)^2 - (4x + 3)(4x - 1)$$

$$= (1)^2 - (3)(-1) = 4$$

Sol.85.(d)

$$\frac{0.01 \times 0.01 \times 0.01 + 0.003 \times 0.003 \times 0.003}{0.05 \times 0.05 - 0.015 \times 0.05 + 0.015 \times 0.015}$$

$$= \frac{1}{25} \times \frac{(0.01)^3 + (0.003)^3}{(0.01)^2 - 0.003 \times 0.01 + (0.003)^2}$$

$$\left\{ \frac{(a)^3 + (b)^3}{(a)^2 - ab + (b)^2} = (a + b) \right\}$$

$$= \frac{1}{25} \times (0.013) = \frac{13}{25} \times 10^{-3}$$

Sol.86.(c) $\frac{3y}{1 + \frac{1}{1 + \frac{y}{1-y}}} = 2$

$$\frac{3y}{1 + \frac{1}{\frac{1-y+y}{1-y}}} = \frac{3y}{1 + \frac{1-y}{1}} = \frac{3y}{2-y} = 2$$

$$\Rightarrow 4 - 2y = 3y \Rightarrow y = \frac{4}{5}$$

Sol.87.(d)

$$= \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}}$$

$$= \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} + \sqrt{3}} \times \frac{\sqrt{2} - \sqrt{3}}{\sqrt{2} - \sqrt{3}}$$

$$= \frac{5 - 2\sqrt{6}}{2 - 3} = 2\sqrt{6} - 5$$

Sol.88.(a) $\frac{a^3 - b^3}{a^2 + a \times b + b^2} = (a - b)$

$$= \frac{120^3 - 100^3}{120^2 + 120 \times 100 + 100^2}$$

$$= (120 - 100) = 20$$

Sol.89.(b) $52 \times 52 - 48 \times 48$

$$\Rightarrow (52 + 48)(52 - 48) \Rightarrow 100 \times 4 = 400$$

Sol.90.(b) $\frac{17}{30}, \frac{7}{15}, \frac{27}{50}, \frac{11}{20}$

Multiply by 300 in each fraction, we get
170, 140, 162, 165

Descending order :-

$$\frac{17}{30} > \frac{11}{20} > \frac{27}{50} > \frac{7}{15}$$

Sol.91.(d) $\sqrt{[(13^2 - 8^2) \times 5 + 4]}$

$$= \sqrt{[(105) \times 5 + 4]} = \sqrt{529} = 23$$

Sol.92.(a)

$$= \frac{14 + 7 - 12 + 15 - 17 + 2 - 3 + 17}{2 + 7 - 9 + 2 - 4 + 22}$$

$$= \frac{21 - 12 + 15 - 17 - 3 + 19}{9 - 9 - 4 + 24}$$

$$= \frac{55 - 32}{20} = \frac{23}{20}$$

Sol.93.(b) $\left(\frac{1}{5} \right) + \left[1\frac{1}{4} - \frac{1}{2} \right] \times \left(\frac{2}{5} \right)$

$$= \frac{1}{5} + \left[\frac{3}{4}\right] \times \left(\frac{2}{5}\right) = \frac{1}{5} + \frac{3}{10} = \frac{5}{10} = 0.5$$

0.5 lies between 0.2 and 0.6

$$\text{Sol.94.(d)} \left(\frac{1}{2} + \frac{3}{4}\right) \div \left(\frac{1}{5} + \frac{3}{10}\right)$$

$$\left(\frac{2+3}{4}\right) \div \left(\frac{2+3}{10}\right) = \left(\frac{5}{4}\right) \div \left(\frac{1}{2}\right) = \frac{5}{2}$$

= 2.5 (lies between 2.3 and 2.6)

$$\text{Sol.95.(d)} 0.5 + 0.55 + 0.555 + 0.5555$$

$$= 5(0.1 + 0.11 + 0.111 + 0.1111)$$

$$= 5(0.4321) = 2.1605$$

$$\text{Sol.96.(c)} 5 \times 4 + (3 + 4)^2 \div 7 \times 4$$

$$= 20 + 49 \div 7 \times 4 = 20 + 7 \times 4 = 48$$

$$\text{Sol.97.(b)}$$

$$\sqrt{[(7^2 + 6^3 + 9^2) \div 2 + 52]}$$

$$= \sqrt{[(49 + 216 + 81) \div 2 + 52]}$$

$$= \sqrt{[346 \div 2 + 52]} = \sqrt{[225]} = 15$$

$$\text{Sol.98.(b)} 74 + 88 \div 8 \times 10 \text{ of } 5 - 90 \div 15 + \{44 - (108 \div 12)\}$$

$$= 74 + \frac{88}{8} \times 50 - \frac{90}{15} + \{44 - (\frac{108}{12})\}$$

$$= 74 + 11 \times 50 - 6 + \{44 - 9\}$$

$$= 74 + 550 - 6 + 35 = 659 - 6 = 653$$

$$\text{Sol.99.(b)} 48 \div 4 + A \times 2 - 8 = 14$$

$$\Rightarrow 12 + A \times 2 - 8 = 14 \Rightarrow A \times 2 - 8 = 2$$

$$\Rightarrow A \times 2 = 10 \Rightarrow A = 5$$

$$\text{Sol.100.(a)}$$

$$\left(\frac{1}{2}\right) \text{ of } 20 + \left(\frac{1}{4}\right) \text{ of } 28 - \left(\frac{1}{3}\right) \text{ of } x = 10$$

$$\Rightarrow 10 + 7 - \left(\frac{x}{3}\right) = 10 \Rightarrow 17 - 10 = \left(\frac{x}{3}\right)$$

$$\Rightarrow x = 7 \times 3 = 21$$

$$\text{Sol.101.(d)}$$

$$\frac{2.5 \times 2.5 + 0.3 \times 0.3 + 5 \times 0.3}{1.6 \times 1.6 + 0.6 \times 0.6 - 1.2 \times 1.6}$$

$$= \frac{(2.5)^2 + (0.3)^2 + 2 \times 2.5 \times 0.3}{(1.6)^2 + (0.6)^2 - 2 \times 1.6 \times 0.6}$$

$$= \frac{(2.5 + 0.3)^2}{(1.6 - 0.6)^2} = \frac{(2.8)^2}{1} = 7.84$$

$$\text{Sol.102.(c)}$$

$$\sqrt{0.52 \times 0.52 + 0.31 \times 0.31 + 0.62 \times 0.52}$$

$$= \sqrt{(0.52)^2 + (0.31)^2 + 2 \times 0.52 \times 0.31}$$

$$= \sqrt{(0.52 + 0.31)^2} = (0.52 + 0.31)$$

$$= 0.83$$

$$\text{Sol.103.(a)} 2 + 4 \div (A + 6) \times 2 = \frac{16}{7}$$

$$\Rightarrow \frac{4}{(A + 6)} \times 2 = \frac{16}{7} - 2 \Rightarrow \frac{8}{(A + 6)} = \frac{2}{7}$$

$$\Rightarrow 56 = 2A + 12 \Rightarrow A = \frac{56 - 12}{2} = 22$$

$$\text{Sol.104.(d)}$$

$$\frac{1.2 \times 1.2 + 0.2 \times 0.2 + 1.2 \times 0.4}{1.8 \times 1.8 + 0.4 \times 0.4 - 1.8 \times 0.8}$$

$$= \frac{(1.2)^2 + (0.2)^2 + 2 \times 1.2 \times 0.2}{(1.8)^2 + (0.4)^2 - 2 \times 1.8 \times 0.4}$$

$$= \frac{(1.2 + 0.2)^2}{(1.8 - 0.4)^2} = \frac{(1.4)^2}{(1.4)^2} = 1$$

$$\text{Sol.105.(d)}$$

$$\text{Formula :- } (a + b)^2 = a^2 + b^2 + 2ab$$

$$\sqrt{0.48 \times 0.48 + 0.25 \times 0.25 + 0.5 \times 0.48}$$

$$= \sqrt{(0.48 + 0.25)^2} = 0.73$$

$$\text{Sol.106.(a)}$$

$$\frac{\frac{49}{63} \div \frac{21}{9} - \frac{85}{510} + \frac{88}{132} \div \frac{66}{11}}{\frac{1}{7} \div \frac{33}{21} + \frac{26}{143} - \frac{1}{209} \div \frac{1}{19}}$$

$$= \frac{\frac{49}{63} \times \frac{9}{21} - \frac{85}{510} + \frac{88}{132} \times \frac{11}{66}}{\frac{1}{7} \times \frac{21}{33} + \frac{26}{143} - \frac{1}{209} \times 19}$$

$$= \frac{\frac{1}{3} - \frac{1}{6} + \frac{1}{9}}{\frac{1}{11} + \frac{2}{11} - \frac{1}{11}} = \frac{6 - 3 + 2}{18} \times \frac{11}{2}$$

$$= \frac{\frac{1}{3} - \frac{1}{6} + \frac{1}{9}}{\frac{1}{11} + \frac{2}{11} - \frac{1}{11}} = \frac{6 - 3 + 2}{18} \times \frac{11}{2}$$

$$= \frac{55}{36}$$

$$\text{Sol.107.(d)}$$

$$\frac{\frac{5}{2} \div \frac{2}{5} \times \frac{1}{5} - \frac{1}{5} \div \frac{3}{5} \times \frac{5}{2}}{\frac{2}{5} \div \frac{1}{5} - \frac{1}{2} \times \frac{5}{2} \div \frac{25}{4} + \frac{1}{2} \div \frac{2}{5} \times \frac{1}{5}}$$

$$= \frac{\frac{25}{4} \times \frac{1}{5} - \frac{1}{3} \times \frac{5}{2}}{2 - \frac{1}{2} \times \frac{2}{5} + \frac{5}{4} \times \frac{1}{5}}$$

$$= \frac{\frac{5}{4} - \frac{5}{6}}{2 - \frac{1}{5} + \frac{1}{4}} = \frac{5}{12} \times \frac{20}{41} = \frac{25}{123}$$

$$= \frac{\frac{5}{4} - \frac{5}{6}}{2 - \frac{1}{5} + \frac{1}{4}} = \frac{5}{12} \times \frac{20}{41} = \frac{25}{123}$$

$$\text{Sol.108.(a)} A = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))} \text{ and}$$

$$B = \frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$$

$$\text{So, } (A + B) = \frac{(36 \div 18 \times 5)}{(36 \div (3 \times 4))} +$$

$$\frac{(5 \div 15 \times 5 + 6 \times 2 - 5)}{(15 \div 5 \times 25 - 2 \times 3 + 6)}$$

$$= \frac{10}{3} + \frac{26}{75} = \frac{75 \times 10 + 26}{225} = \frac{776}{225}$$

$$\text{Sol.109.(b)} \text{ We can write the Fraction}$$

$$\frac{5}{11}, \frac{5}{7}, \frac{3}{8}, \frac{6}{13} \text{ as } \frac{30}{66}, \frac{30}{42}, \frac{30}{80}, \frac{30}{65}$$

respectively

So, the smallest and greatest fraction

$$\text{respectively are } \frac{30}{80} \left(\frac{3}{8}\right) \text{ and } \frac{30}{42} \left(\frac{5}{7}\right)$$

Therefore, difference of smallest and

$$\text{largest fractions} = \frac{5}{7} - \frac{3}{8} = \frac{19}{56}$$

$$\text{Sol.110.(a)} \text{ from the formula,}$$

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$\frac{(0.91 + 0.09)(0.91)^2 + (0.09)^2 - .0819}{[(0.91)^2 - .0819 + (0.09)^2]} = 1$$

$$\text{Sol.111.(a)}$$

$$P = \frac{1}{4} \text{ of } \frac{4}{5} + \frac{9}{2} \text{ of } \frac{17}{3} - \frac{3}{10}$$

$$= \frac{1}{4} \times \frac{4}{5} + \frac{9}{2} \times \frac{17}{3} - \frac{3}{10}$$

$$= \frac{4}{20} + \frac{51}{2} - \frac{3}{10}$$

$$= \frac{4 + 510 - 6}{20} = \frac{127}{5}$$

$$\text{Sol.112.(d)} A = \frac{18 \div 9 \times 4}{15 \div 3 \times 5} = \frac{8}{25}$$

$$B = \frac{18 \div 36 \times 3 + 5 \times 1 - 6}{18 \div 6 \times 20 - 3 \times 4 + 12}$$

$$= \frac{\frac{1}{2} \times 3 - 1}{60 - 12 + 12} = \frac{1}{120}$$

$$A + B = \frac{8}{25} + \frac{1}{120} = \frac{192 + 5}{600} = \frac{197}{600}$$

$$\text{Sol.113.(a)}$$

$$\sqrt{a - \sqrt{a - \sqrt{a - \sqrt{a - \dots \infty}}}}$$

$$= \frac{\sqrt{4a + 1} - 1}{2}$$

$$\sqrt{20 - \sqrt{20 - \sqrt{20 - \sqrt{20 - \dots \infty}}}}$$

$$= \frac{\sqrt{4 \times 20 + 1} - 1}{2} = \frac{\sqrt{81} - 1}{2} = \frac{9 - 1}{2} = 4$$

Short Trick:-

Co - prime factors of 20 = (4, 5)

In case of (-ve) sign, smaller number is the right answer.

$$\text{Sol.114.(b)}$$

$$\sqrt{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}}$$

$$= \frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}}{\sqrt{16 + 19.25 \times 4^2}}$$

$$= \frac{\sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}}{\sqrt{324}}$$

$$= \frac{\sqrt{10 + \sqrt{25 + 11}}}{18} = \frac{\sqrt{10 + 6}}{18} = \frac{2}{9}$$

$$\text{Sol.115.(a)} x^9 \times x^5 \times x^{-4} \times x^0 \times x^{-6}$$

$$= x^9 \times x^5 \times \frac{1}{x^4} \times 1 \times \frac{1}{x^6} \dots \dots \dots (x^0 = 1)$$

$$= \frac{x^{14}}{x^{10}} = x^4 \dots \dots \dots \left(\text{as } \frac{p^m}{p^n} = p^{m-n}\right)$$

$$\text{Sol.116.(c)} \text{ Sum of 'n' natural numbers}$$

$$= \frac{n}{2}(a + l) \dots \text{where } a = \text{first term,}$$

l = last term

$$100^2 - 99^2 + 98^2 - 97^2 + \dots + 12^2 - 11^2$$

$$(100 + 99)(100 - 99) + (98 + 97)$$

$$(98 - 97) + \dots + (12 + 11)(12 - 11)$$

$$= 100 + 99 + 98 + 97 \dots \dots \dots + 12 + 11$$

$$\text{Sum} = \frac{90}{2}(100 + 11) = 4995$$

$$\text{Sol.117.(c)}$$

$$\text{Formula :- } a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$(5x + 8y)(25x^2 + 64y^2 - 40xy)$$

$$= (5x + 8y)\{(5x)^2 + (8y)^2 - 5x \times 8y\}$$

$$= (5x)^3 + (8y)^3 = 125x^3 + 512y^3$$

Sol.118.(a) $\sqrt{36x^2 - 108x + 81}$

$$= \sqrt{(6x - 9)^2} = 6x - 9$$

Sol.119.(c) $\frac{\sqrt[3]{-2744} \times \sqrt[3]{-216}}{\sqrt[3]{\frac{64}{729}}}$

$$= \frac{(-14) \times (-6)}{\frac{4}{9}} = \frac{14 \times 6 \times 9}{4} = 189$$

Sol.120.(d) $81^{\frac{3}{4}} + [(20 \div 5 \text{ of } 3 \times 6) + \{(8 \div 24 \text{ of } 3) \times 4\} - 10 \div 5] - (\frac{1}{32})^{-\frac{2}{5}}$

$$= (3^4)^{\frac{3}{4}} + [(20 \div 15 \times 6) + \{(8 \div 72) \times 4\} - 2] - (\frac{1}{2})^{5 \times \frac{-2}{5}}$$

$$= 27 + [8 + \frac{4}{9} - 2] - 4$$

$$= 27 + \frac{58}{9} - 4 = 23 + \frac{58}{9}$$

$$= 23 + 6\frac{4}{9} = 29\frac{4}{9}$$

Sol.121.(b) $\{(5^6)^5 \times (5^5)^8\} \div 25 = (625)^a$

$$5^{30+40} \div 5^2 = (625)^a$$

$$5^{70} \div 5^2 = (625)^a \Rightarrow 5^{68} = (625)^a$$

$$(5^4)^{17} = (625)^a \Rightarrow 625^{17} = (625)^a \Rightarrow a = 17$$

Sol.122.(a) $(7^3 - 7^2 + 6)(6^3 - 2^3 - 3^2 + 1) \div (21^2 + 20^2 - 41)$

$$\Rightarrow (343 - 49 + 6)(216 - 8 - 9 + 1) \div (441 + 400 - 41)$$

$$\Rightarrow 300 \times 200 \div 800 = 75$$

Sol.123.(b) $\sqrt{36 \div 15 \text{ of } 2 \text{ of } [25 \times 4 \div 4 \text{ of } \{29 + (12 - 11)\}]}$

$$= \sqrt{36 \div 15 \text{ of } 2 \text{ of } [25 \times 4 \div 4 \text{ of } \{29 + 1\}]}$$

$$= \sqrt{36 \div 15 \text{ of } 2 \text{ of } [25 \times 4 \div 120]}$$

$$= \sqrt{36 \div 15 \text{ of } 2 \text{ of } \frac{5}{6}} = \sqrt{36 \div 25}$$

$$= \sqrt{\frac{36}{25}} = \frac{6}{5} = 1\frac{1}{5}$$

Sol.124.(b) $99\frac{11}{99} + 99\frac{13}{99} + 99\frac{15}{99} + \dots + 99\frac{67}{99}$

$$= (99 + \frac{11}{99}) + (99 + \frac{13}{99}) + (99 + \frac{15}{99}) + \dots + (99 + \frac{67}{99})$$

$$= 99 \times 29 + \frac{1}{99} (11 + 13 + 15 + \dots + 67)$$

$$= 2871 + \frac{1}{99} (29 \times \frac{67+11}{2})$$

$$= 2871 + \frac{1}{99} (29 \times 39)$$

$$= 2871 + \frac{377}{33} = \frac{94743 + 377}{33} = \frac{95120}{33}$$

Sol.125.(c) Let $x = y = z = 2$

$$\frac{(x+y+z)(xy+yz+zx) - xyz}{(x+y)(y+z)(z+x)}$$

$$= \frac{(2+2+2)(2 \times 2 + 2 \times 2 + 2 \times 2) - 2 \times 2 \times 2}{(2+2)(2+2)(2+2)}$$

$$= \frac{(6)(4+4+4) - 8}{(4)(4)(4)} = \frac{(6)(12) - 8}{64}$$

$$= \frac{72 - 8}{64} = \frac{64}{64} = 1$$

Sol.126.(b)

$$\frac{\sqrt{29.16}}{\sqrt{1.1664}} + \frac{\sqrt{0.2916}}{\sqrt{116.64}} + \frac{\sqrt{0.0036}}{\sqrt{0.36}}$$

$$= \frac{5.4}{1.08} + \frac{0.54}{10.8} + \frac{0.06}{0.6}$$

$$= 5 + \frac{1}{20} + \frac{1}{10} = 5 + \frac{3}{20} = \frac{103}{20}$$

Sol.127.(c)

$$A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$$

$$= \frac{0.02 \times 0.02}{2 \times 10^{\frac{3}{4}} \times 50 \times 3 \times 10^{\frac{1}{4}}}$$

$$= \frac{0.0004}{3 \times 10^{2 + \frac{3}{4} + \frac{1}{4}}} = \frac{0.0004}{3 \times 10^3} = \frac{4}{3} \times 10^{-7}$$

$$B = \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}} = \frac{0.9 \times 0.2}{0.4}$$

$$= \frac{18}{40} = \frac{9}{20}$$

Now,

$$A \times B = \frac{4}{3} \times 10^{-7} \times \frac{9}{20} = \frac{3}{5} \times 10^{-7}$$

Sol.128.(a)

$$A = 0.3\overline{12} = \frac{312 - 3}{990} = \frac{309}{990} = \frac{103}{330}$$

$$B = 0.4\overline{15} = \frac{415 - 4}{990} = \frac{411}{990} = \frac{137}{330}$$

$$C = 0.30\overline{9} = \frac{309 - 30}{900} = \frac{279}{900} = \frac{31}{100}$$

So, $A + B + C = \frac{103}{330} + \frac{137}{330} + \frac{31}{100}$

$$= \frac{240}{330} + \frac{31}{100} = \frac{80}{110} + \frac{31}{100} = \frac{1141}{1100}$$

Sol.129.(a)

$$\sqrt{\frac{\sqrt{5}-1}{\sqrt{5}+1}} = \sqrt{\frac{(\sqrt{5}-1)(\sqrt{5}-1)}{(\sqrt{5}+1)(\sqrt{5}-1)}}$$

$$= \sqrt{\frac{(\sqrt{5}-1)^2}{4}} = \frac{(\sqrt{5}-1)}{2} = \frac{2.24-1}{2}$$

$$= \frac{1.24}{2} = 0.62$$

Sol.130.(d)

If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$

$$\Rightarrow 13^3 + (-8)^3 + (-5)^3 = 13 \times 8 \times 5 = 0$$

$$\Rightarrow 13^3 + (-8)^3 + (-5)^3$$

$$= 3 \times 13 \times (-8) \times (-5) = 1560$$

Sol.131.(c)

$$(0.064)^{123} \div 0.16^{47} \times 0.4^{34} \times 0.4^{29} = 0.4^x$$

$$\Rightarrow 0.4^x = 0.4^{3 \times 123} \div 0.4^{47 \times 2} \times 0.4^{34} \times 0.4^{29}$$

$$\Rightarrow 0.4^x = 0.4^{369-94} \times 0.4^{34} \times 0.4^{29}$$

$$\Rightarrow 0.4^x = 0.4^{275} \times 0.4^{34} \times 0.4^{29}$$

$$\Rightarrow 0.4^x = 0.4^{275+34+29} \Rightarrow 0.4^x = 0.4^{338}$$

We know that, If bases are equal then powers will be equal.

Therefore, $x = 338$

Sol.132.(b) $\frac{2x+3}{18 \div 3 \times 4 + 2} = \frac{12 \times 3 + 5}{2 \times 4 \div 2}$

$$\Rightarrow \frac{2x+3}{6 \times 4 + 2} = \frac{36+5}{2 \times 2} \Rightarrow \frac{2x+3}{26} = \frac{41}{4}$$

$$\Rightarrow 8x + 12 = 1066 \Rightarrow 8x = 1054$$

$$\Rightarrow x = 131.75$$

Sol.133.(a) $A + B = (\frac{2}{3} + \frac{4}{7} \text{ of } \frac{14}{12}) + (\frac{5}{12} \text{ of } \frac{7}{15} \times \frac{36}{21})$

$$= \frac{2}{3} + \frac{2}{3} + \frac{1}{3} = \frac{5}{3}$$

Sol.134.(d)

$$\frac{\frac{5}{6} - \frac{5}{11} \text{ of } [\frac{3}{7} + \frac{8}{12} \div \frac{6}{24} + \frac{5}{7}]}{\frac{\frac{3}{12} \times \frac{6}{5}}{\frac{5}{6} - \frac{5}{11} \times \frac{80}{21}} \Rightarrow \frac{\frac{385-800}{3 \times 2 \times 77}}{\frac{3}{10}}}$$

$$= \frac{-415 \times 10}{3 \times 3 \times 2 \times 77} = \frac{-2075}{693}$$

Sol.135.(d) $P = 7 \times 4 - 3 = 25$

$$Q = 8 \div 4 + 2 = 4, R = 14 \div 21 \times 9 = 6$$

$$P \times Q + R = 25 \times 4 + 6 = 106$$

Sol.136.(c) $(x-y)(x+y) = x^2 - y^2$

$$\Rightarrow (36p^2 + 49q^2) \{(6p)^2 - (7q)^2\}$$

$$\Rightarrow (36p^2 + 49q^2)(36p^2 - 49q^2)$$

$$\Rightarrow (36p^2)^2 - (49q^2)^2 = 1296p^4 - 2401q^4$$

Sol.137.(d)

$$3p - [3p - \overline{p+q} - \{3p - (p - \overline{q-p})\}]$$

$$= 3p - [3p - \overline{p+q} - \{3p - (p - q + p)\}]$$

$$= 3p - [3p - \overline{p+q} - (3p - 2p + q)]$$

$$= 3p - [3p - \overline{p+q} - p - q]$$

$$= 3p - [3p - p - q - p - q]$$

$$= 3p - p + 2q = 2p + 2q$$

Sol.138.(c) $[7\frac{1}{2} \div \{1\frac{1}{4} - \frac{1}{2}(2\frac{1}{2} - \frac{1}{4} - \frac{1}{6})\}] \div (\frac{1}{2} \text{ of } 8\frac{1}{3})$

$$= [\frac{15}{2} \div \{\frac{5}{4} - \frac{1}{2}(\frac{5}{2} - \frac{1}{12})\}] \div \frac{25}{6}$$

$$= [\frac{15}{2} \div \{\frac{5}{4} - \frac{1}{2} \times \frac{29}{12}\}] \div \frac{25}{6}$$

$$= [\frac{15}{2} \div \{\frac{5}{4} - \frac{29}{24}\}] \div \frac{25}{6}$$

$$= [\frac{15}{2} \div \frac{1}{24}] \div \frac{25}{6}$$

$$= 180 \times \frac{6}{25} \Rightarrow \frac{216}{5} = 43.2$$

Sol.139.(c)

$$\begin{aligned} & \{(28 - 3^3) + 99\} + 136 \div 4 \times 5 - (44 - 2^2) \\ & = \{(28 - 27) + 99\} + 136 \div 4 \times 5 - (44 - 4) \\ & = \{1 + 99\} + 34 \times 5 - (40) \\ & = 100 + 170 - 40 = 270 - 40 = 230 \end{aligned}$$

$$\text{Sol.140.(a)} \left(\frac{9^3 + 9^4 + 9^5 + 9^6 + 9^7}{9^1 + 9^2 + 9^3 + 9^4 + 9^5} \right)^{\frac{1}{2}}$$

$$\begin{aligned} & \Rightarrow \left[\frac{9^3(1 + 9 + 9^2 + 9^3 + 9^4)}{9(1 + 9 + 9^2 + 9^3 + 9^4)} \right]^{\frac{1}{2}} \\ & \Rightarrow (9^2)^{\frac{1}{2}} = 9 \end{aligned}$$

Sol.141.(b)

$$\begin{aligned} & 3\frac{6}{7} - \{5\frac{1}{7} - (2\frac{3}{7} - 3\frac{5}{7} - 4)\} \\ & = \frac{27}{7} - \left\{ \frac{36}{7} - \left(\frac{17}{7} - \frac{26}{7} - 4 \right) \right\} \\ & = \frac{27}{7} - \left\{ \frac{36}{7} - \left(\frac{17}{7} - \frac{26}{7} + 4 \right) \right\} \\ & = \frac{27}{7} - \left\{ \frac{36}{7} - \left(-\frac{9}{7} + 4 \right) \right\} \\ & = \frac{27}{7} - \left\{ \frac{36}{7} - \left(\frac{19}{7} \right) \right\} = \frac{27}{7} - \left\{ \frac{17}{7} \right\} = \frac{10}{7} \end{aligned}$$

$$\text{Sol.142.(c)} \sqrt{a^{-1}b} \times \sqrt{b^{-1}c} \times \sqrt{c^{-1}a}$$

$$\begin{aligned} & = \sqrt{\frac{b}{a}} \times \sqrt{\frac{c}{b}} \times \sqrt{\frac{a}{c}} \\ & = \sqrt{\frac{b}{a} \times \frac{c}{b} \times \frac{a}{c}} = 1 \end{aligned}$$

$$\begin{aligned} \text{Sol.143.(c)} & 2 \cdot 19^2 + 2(2 \cdot 19)(3 \cdot 81) + 3 \cdot 81^2 \\ & = (2 \cdot 19 + 3 \cdot 81)^2 = 6^2 = 36 \end{aligned}$$

Sol.144.(d) Formula used :-

$$(x^3 + y^3 + z^3) = 3xyz \text{ (when } x + y + z = 0)$$

$$\begin{aligned} & \frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3} \\ & = \frac{3(a-b)(b-c)(c-a)}{3(a^2-b^2)(b^2-c^2)(c^2-a^2)} \\ & = \frac{3(a-b)(b-c)(c-a)}{3(a+b)(a-b)(b+c)(b-c)(c+a)(c-a)} \\ & = \frac{1}{(a+b)(b+c)(c+a)} \end{aligned}$$

Sol.145.(c)

$$\text{Formula used : } (a^3 + b^3 + c^3 - 3abc)$$

$$\begin{aligned} & = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) \\ & \frac{(a+b+c)(a^2+b^2+c^2-ab-bc-ca)}{a^2+b^2+c^2-ab-bc-ca} \\ & \Rightarrow 2.01 + 39.75 + 28.24 = 70 \end{aligned}$$

$$\text{Sol.146.(a)} \sqrt{2 + \sqrt{3}} + \sqrt{2 - \sqrt{3}}$$

$$\begin{aligned} & = \sqrt{\frac{2(2+\sqrt{3})}{2}} + \sqrt{\frac{2(2-\sqrt{3})}{2}} \\ & = \sqrt{\frac{4+2\sqrt{3}}{2}} + \sqrt{\frac{4-2\sqrt{3}}{2}} \\ & = \sqrt{\frac{(\sqrt{3}+1)^2}{2}} + \sqrt{\frac{(\sqrt{3}-1)^2}{2}} \end{aligned}$$

$$= \frac{\sqrt{3}+1}{\sqrt{2}} + \frac{\sqrt{3}-1}{\sqrt{2}} = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{6}$$

Sol.147.(b) Formula used :-

$$a^3 + b^3 + c^3 = 3abc \text{ when } a + b + c = 0$$

$$\begin{aligned} & \frac{(r-s)^3 + (s-t)^3 + (t-r)^3}{6(r-s)(s-t)(t-r)} \\ & = \frac{3(r-s)(s-t)(t-r)}{6(r-s)(s-t)(t-r)} = \frac{3}{6} = \frac{1}{2} \end{aligned}$$

$$\text{Sol.148.(d)} (a^{-1} + b^{-1}) \div (a^{-3} + b^{-3})$$

$$\begin{aligned} & = \left(\frac{1}{a} + \frac{1}{b} \right) \div \left(\frac{1}{a^3} + \frac{1}{b^3} \right) \\ & = \left(\frac{a+b}{ab} \right) \div \left(\frac{a^3+b^3}{a^3b^3} \right) \\ & = \left(\frac{a+b}{ab} \right) \times \left(\frac{a^3b^3}{a^3+b^3} \right) = \frac{a^2b^2}{a^2-ab+b^2} \end{aligned}$$

Sol.149.(c) Let the required number be y.

$$\begin{aligned} & \text{According to the question,} \\ & = (5 \times 5 \times 5 \times 5 \times 5)^5 \times (5 \times 5 \times 5)^5 \div 5 = (125)^y \\ & = (5^5)^5 \times (5^3)^5 \div 5 = (125)^y \\ & = 5^{25} \times 5^{15} \div 5 = (5^3)^y \\ & = 5^{25+15} \div 5 = 5^{3y} \Rightarrow 5^{40} \div 5 = 5^{3y} \\ & = 5^{40-1} = 5^{3y} \Rightarrow 5^{39} = 5^{3y} \\ & = 39 = 3y \Rightarrow y = 13 \end{aligned}$$

Sol.150.(b) Put a = b = 1 and c = 0.

$$\begin{aligned} & \frac{(a^2-4b^2)^3 + 64(b^2-4c^2)^3 + (16c^2-a^2)^3}{(a-2b)^3 + (2b-4c)^3 + (4c-a)^3} \\ & \frac{(1^2-4 \times 1^2)^3 + 64(1^2-4 \times 0^2)^3 + (16 \times 0^2-1^2)^3}{(1-2)^3 + (2-4 \times 0)^3 + (4 \times 0-1)^3} \\ & = \frac{36}{6} = 6 \end{aligned}$$

On checking the option we get only option (b) satisfies this.

Sol.151.(c) Given that

$$\begin{aligned} & 5\sqrt{3} + \sqrt{75} = 17.32 \\ & 5\sqrt{3} + 5\sqrt{3} = 10\sqrt{3} = 17.32 \\ & \text{Now according to question} \\ & 14\sqrt{3} + \sqrt{108} = 14\sqrt{3} + 6\sqrt{3} = 20\sqrt{3} \\ & 2 \times 17.32 = 34.64 \end{aligned}$$

Sol.152.(a) $847 = 7^1 \times 11^2$, $385 = 5 \times 7 \times 11$

$$\begin{aligned} & 675 = 3^3 \times 5^2, 3025 = 5^2 \times 11^2 \\ & 847 \times 385 \times 675 \times 3025 = 3^a \times 5^b \times 7^c \times 11^d \\ & 3^3 \times 5^5 \times 7^2 \times 11^5 = 3^a \times 5^b \times 7^c \times 11^d \end{aligned}$$

On comparing, we get

$$a = 3, b = 5, c = 2, d = 5$$

$$ab - cd = 3 \times 5 - 2 \times 5 = 5$$

Sol.153.(a)

$$\begin{aligned} & \frac{1}{4 - \sqrt{15}} = \frac{(4 + \sqrt{15})}{(4 - \sqrt{15})(4 + \sqrt{15})} \\ & = 4 + \sqrt{15} \end{aligned}$$

Similarly, we get conjugate of all values,

$$\begin{aligned} & 4 + \sqrt{15} - \sqrt{15} - \sqrt{14} + \sqrt{14} + \sqrt{13} - \sqrt{13} - \sqrt{12} \\ & + \sqrt{12} + \sqrt{11} - \sqrt{11} - \sqrt{10} + \sqrt{10} + 3 - 3 - \sqrt{8} \end{aligned}$$

$$\Rightarrow 4 - \sqrt{8} = 4 - 2\sqrt{2}$$

$$\text{Sol.154.(c)} \frac{\sqrt{38-5\sqrt{3}}}{\sqrt{26+7\sqrt{3}}}$$

on multiplying numerator and denominator by $\sqrt{2}$ we get,

$$\frac{\sqrt{76-10\sqrt{3}}}{\sqrt{52+14\sqrt{3}}} = \frac{\sqrt{(5\sqrt{3}-1)^2}}{\sqrt{(7+\sqrt{3})^2}} = \frac{5\sqrt{3}-1}{7+\sqrt{3}}$$

$$\frac{(5\sqrt{3}-1)(7-\sqrt{3})}{(7+\sqrt{3})(7-\sqrt{3})} = \frac{-11+18\sqrt{3}}{23}$$

On comparing $\frac{-11+18\sqrt{3}}{23}$ with

$$\frac{a+b\sqrt{3}}{23} \text{ we get, } \Rightarrow a = -11 \text{ and } b = 18$$

$$(b-a) = 18 - (-11) = 29$$

Sol.155.(d) $\frac{3}{8} + \frac{1}{7} \times 1\frac{3}{4} + (\frac{2}{3} \div \frac{3}{10})$ of

$$\begin{aligned} & \left(\frac{4}{9} \right) - \left(3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3} \right) \times \frac{15}{8} \\ & = \frac{3}{8} + \frac{1}{7} \times \frac{7}{4} + \left(\frac{2}{3} \div \frac{3}{10} \times \frac{4}{9} \right) - \\ & \quad \left(\frac{16}{5} \div \frac{9}{2} \text{ of } \frac{16}{3} \right) \times \frac{15}{8} \\ & = \frac{3}{8} + \frac{1}{4} + \left(\frac{2}{3} \div \frac{2}{15} \right) - \left(\frac{16}{5} \div 24 \right) \times \frac{15}{8} \\ & = \frac{3}{8} + \frac{1}{4} + \left(\frac{2}{3} \times \frac{15}{2} \right) - \left(\frac{16}{5} \times \frac{1}{24} \right) \times \frac{15}{8} \\ & = \frac{3}{8} + \frac{1}{4} + (5) - \left(\frac{2}{15} \right) \times \frac{15}{8} \\ & = \frac{3}{8} + \frac{1}{4} + 5 - \frac{1}{4} \Rightarrow \frac{3}{8} + 5 = 5\frac{3}{8} \end{aligned}$$

$$\text{Sol.156.(a)} \left\{ 4 - \frac{2}{1 + \frac{2}{1 - \frac{1}{2 + \frac{3}{4}}}} \right\} \div 1\frac{5}{12}$$

$$\text{of } \frac{72}{145} - (4 + 3 \div 0.5 - 1)$$

$$= \left\{ 4 - \frac{2}{1 + \frac{2}{1 - \frac{4}{11}}} \right\} \div \frac{17}{12} \times \frac{72}{45} -$$

$$\begin{aligned} & (4 + 6 - 1) \\ & = \left\{ 4 - \frac{2}{1 + \frac{22}{7}} \right\} \div \frac{102}{145} - (9) \end{aligned}$$

$$= \left\{ 4 - \frac{14}{29} \right\} \div \frac{102}{145} - (9)$$

$$= \left\{ \frac{102}{29} \right\} \div \frac{102}{145} - (9)$$

$$= \left\{ \frac{102}{29} \right\} \times \frac{145}{102} - (9) = 5 - 9 = -4$$

Sol.157.(d) $4\frac{2}{5} \div \{(1\frac{1}{2} - 3\frac{1}{5}) \div 3\frac{2}{5}$

$$+ (2\frac{1}{5} \div 1\frac{1}{2} + 4\frac{2}{5}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{(\frac{3}{2} - \frac{16}{5}) \div \frac{17}{5} +$$

$$(\frac{11}{5} \div \frac{3}{2} + \frac{22}{5}) + \frac{1}{2}\}$$

$$= \frac{22}{5} \div \{(-\frac{17}{10}) \times \frac{5}{17} +$$

$$\begin{aligned} & \left(\frac{11}{5} \times \frac{2}{3} + \frac{22}{5} \right) + \frac{1}{2} \\ &= \frac{22}{5} \div \left\{ -\frac{1}{2} + \left(\frac{88}{15} \right) + \frac{1}{2} \right\} \\ &= \frac{22}{5} \div \left\{ \frac{88}{15} \right\} = \frac{22}{5} \times \frac{15}{88} = \frac{3}{4} \end{aligned}$$

Sol.158.(a)

$$\begin{aligned} & 3\frac{5}{6} + \left[3\frac{2}{3} + \left\{ \frac{15}{4} \left(5\frac{4}{5} \div 14\frac{1}{2} \right) \right\} \right] \\ &= \frac{23}{6} + \left[\frac{11}{3} + \left\{ \frac{15}{4} \left(-\frac{29}{5} \div \frac{29}{2} \right) \right\} \right] \\ &= \frac{23}{6} + \left[\frac{11}{3} + \left\{ \frac{15}{4} \times \frac{2}{5} \right\} \right] \\ &= \frac{23}{6} + \left[\frac{11}{3} + \frac{3}{2} \right] = \frac{23}{6} + \frac{31}{6} = 9 \end{aligned}$$

Sol.159.(c)

$$\begin{aligned} & \left(\frac{3}{4} - \frac{1}{4} \div \frac{1}{4} \text{ of } \frac{2}{5} \right) \div \left(\frac{3}{4} \div \frac{2}{3} \text{ of } \frac{3}{5} \right) \\ &= \left(\frac{3}{4} - \frac{1}{4} \div \frac{1}{10} \right) \div \left(\frac{3}{4} \div \frac{2}{5} \right) \\ &= -\frac{7}{4} \div \left(-\frac{15}{8} \right) = -\frac{14}{15} \end{aligned}$$

Sol.160.(b) $(2x - 3y)^3 - 18xy(2x - 3y)$
 $\Rightarrow 8x^3 - 27y^3 - 36x^2y + 54xy^2 - 36x^2y + 54xy^2$
 $\Rightarrow 8x^3 - 27y^3 - 72x^2y + 108xy^2$

Sol.161.(a)

$$\begin{aligned} & 364 - [142 - \{75 + (38 - \frac{5}{4} + \frac{14}{8})\}] \\ & 364 - [142 - \{75 + (38 - 3)\}] = 332 \end{aligned}$$

Sol.162.(c) $2.\bar{4} = 2\frac{4}{9} = \frac{22}{9}$

$$0.\bar{6} = \frac{2}{3}, \quad 0.1\bar{6} = \frac{1}{6}$$

$$0.\bar{27} = \frac{3}{11}, \quad 0.8\bar{3} = \frac{5}{6}$$

$$\begin{aligned} & \Rightarrow (2.\bar{4} \times 0.\bar{6} \times 3 \times 0.1\bar{6}) \times \\ & \quad [0.\bar{27} \times (0.8\bar{3} \div 0.1\bar{6})] \\ & \Rightarrow \left(\frac{22}{9} \times \frac{2}{3} \times 3 \times \frac{1}{6} \right) \times \left[\frac{3}{11} \times \frac{5}{6} \times \frac{6}{1} \right] \\ & \Rightarrow \left(\frac{22}{27} \right) \times \left(\frac{15}{11} \right) \Rightarrow \frac{10}{9} = 1\frac{1}{9} = 1.\bar{1} \end{aligned}$$

Sol.163.(c) $\frac{15(\sqrt{10} + \sqrt{5})}{\sqrt{10} + \sqrt{20} + \sqrt{40} - \sqrt{5} - \sqrt{80}}$
 $= \frac{15\sqrt{5}(\sqrt{2} + 1)}{\sqrt{5}(\sqrt{2} + 2 + 2\sqrt{2} - 1 - 4)} = \frac{15(\sqrt{2} + 1)}{(3\sqrt{2} - 3)}$
 $= \frac{5(\sqrt{2} + 1)}{(\sqrt{2} - 1)} \text{ (on rationalizing)}$
 $= 5(3 + 2\sqrt{2})$

Sol.164.(d) $\sqrt{11 - 3\sqrt{8}} = a + b\sqrt{2}$,

$$\sqrt{9 + 2 - 6\sqrt{2}} = a + b\sqrt{2},$$

$$3 - \sqrt{2} = a + b\sqrt{2},$$

$$a = 3, b = -1 \Rightarrow 2a + 3b = 6 - 3 = 3$$

Sol.165.(a) $\frac{7+3\sqrt{5}}{3+\sqrt{5}} - \frac{7-3\sqrt{5}}{3-\sqrt{5}}$
 $\Rightarrow \frac{(7+3\sqrt{5})(3-\sqrt{5}) - (7-3\sqrt{5})(3+\sqrt{5})}{(3)^2 - (\sqrt{5})^2}$

$$\Rightarrow \frac{18\sqrt{5} - 14\sqrt{5}}{4} = \frac{4\sqrt{5}}{4} = \sqrt{5} = 2.2360$$

So, 2.236 lies between 2 and 2.5

Sol.166.(a)

$$\begin{aligned} & \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{7} + 4\sqrt{3}}} \\ & \Rightarrow \sqrt{-\sqrt{3} + \sqrt{3 + 8\sqrt{(2 + \sqrt{3})^2}}} \\ & \Rightarrow \sqrt{-\sqrt{3} + \sqrt{19 + 8\sqrt{3}}} \\ & \Rightarrow \sqrt{-\sqrt{3} + \sqrt{(4 + \sqrt{3})^2}} = \sqrt{4} = 2 \end{aligned}$$

Sol.167.(b) $5 - \frac{8+2\sqrt{15}}{4} - \frac{1}{8+2\sqrt{15}}$

on rationalisation

$$\begin{aligned} & 5 - \frac{8+2\sqrt{15}}{4} - \frac{1}{8+2\sqrt{15}} \times \frac{8-2\sqrt{15}}{8-2\sqrt{15}} \\ & \Rightarrow 5 - \frac{8+2\sqrt{15}}{4} - \frac{8-2\sqrt{15}}{4} \Rightarrow 5 - 4 = 1 \end{aligned}$$

Sol.168.(c) $\sqrt[3]{15625} - \sqrt{x} = 4$

$$\Rightarrow 25 - \sqrt{x} = 4 \Rightarrow 21 = \sqrt{x} \Rightarrow x = 441$$

Sol.169.(a) $\left(\frac{2}{5} \text{ of } 6\frac{1}{4} \div \frac{3}{7} \right) \text{ of } 1\frac{2}{7}$

$$x = \left[\left(\frac{2}{5} \text{ of } 6\frac{1}{4} \div \frac{3}{7} \right) \text{ of } 1\frac{2}{7} \right]^2$$

$$x = \left[\left(\frac{2}{5} \text{ of } 6\frac{1}{4} \div \frac{3}{7} \right) \text{ of } 1\frac{2}{7} \right]^2$$

$$x = \left[\left(\frac{2}{5} \times \frac{25}{4} \div \frac{3}{7} \right) \times \frac{9}{7} \right]^2 \div \frac{45}{4}$$

$$x = \left[\left(\frac{2}{5} \times \frac{25}{4} \times \frac{7}{3} \right) \times \frac{9}{7} \right]^2 \times \frac{4}{45}$$

$$x = \left[\frac{2}{3} \right]^2 = \frac{4}{9} \Rightarrow 81x = 81 \times \frac{4}{9} = 36$$

Sol.170.(b)

$$\frac{5\frac{1}{2} \div 3\frac{2}{3} \text{ of } \frac{1}{4} + (5\frac{1}{9} - 7\frac{7}{8} \div 9\frac{9}{20}) \times \frac{9}{11}}{5 \div 5 \text{ of } \frac{1}{10} - 10 \times 10 \div 20}$$

$$= \frac{5\frac{1}{2} \div \frac{11}{12} + \left(\frac{77}{18} \right) \times \frac{9}{11}}{5 \div \frac{1}{2} - 10 \times 10 \div 20}$$

$$= \frac{6 + \frac{7}{2}}{10 - 5} = 1\frac{9}{10}$$

Sol.171.(b)

Simplifying the numerator firstly,

$$= \frac{1}{3} + \left[4\frac{3}{4} - \left(\frac{19}{6} - \frac{14}{6} \right) \right]$$

$$= \frac{1}{3} + \left[4\frac{3}{4} - \frac{5}{6} \right] = \frac{1}{3} + \left[\frac{19}{4} - \frac{5}{6} \right] = \frac{51}{12}$$

Now Simplifying the denominator

$$\left(\frac{1}{5} \text{ of } \frac{1}{5} \div \frac{1}{5} \right) \div \left(\frac{1}{5} \div \frac{1}{5} \times \frac{1}{5} \right)$$

$$\frac{1}{5} \div \frac{1}{5} = 1$$

On combining the two simplified parts

$$= \frac{51}{12} = 4.25$$

So the value 4.25 lies between 4.2 and 4.4

Sol.172.(c)

$$\sqrt{1875} = 25\sqrt{3}, \quad \sqrt{3888} = 36\sqrt{3}$$

$$\sqrt{1200} = 20\sqrt{3}, \quad \sqrt{768} = 16\sqrt{3}$$

$$\sqrt{175} = 5\sqrt{7}, \quad \sqrt{1792} = 16\sqrt{7}$$

$$\frac{25\sqrt{3}}{36\sqrt{3}} \times \frac{16\sqrt{3}}{20\sqrt{3}} \times \frac{5\sqrt{7}}{16\sqrt{7}} = \frac{25}{144}$$

$$\sqrt{x} = \frac{5}{12}$$

Sol.173.(d)

$$\left(2\frac{1}{3} + 2\frac{1}{2} - \frac{1}{6} \right) \div 2\frac{1}{3} \times 5\frac{2}{3} \div 1\frac{2}{3} \text{ of } 4\frac{1}{4}$$

$$3\frac{1}{5} \div 4\frac{1}{2} \text{ of } 5\frac{1}{3} + 5\frac{1}{3} \times \frac{3}{4} \div 2\frac{2}{3}$$

Simplifying the numerator

$$= \frac{14}{3} \times \frac{3}{7} \times \frac{17}{3} \times \frac{12}{85} = \frac{8}{5}$$

Simplifying the Denominator

$$\frac{16}{5} \div \frac{9}{2} \text{ of } \frac{16}{3} + \frac{16}{3} \times \frac{3}{4} \div \frac{8}{3}$$

$$= \frac{16}{5 \times 24} + \frac{16}{3} \times \frac{9}{32}$$

$$= \frac{2}{15} + \frac{3}{2} = \frac{49}{30}$$

Combining the simplified forms

$$= \frac{\frac{8}{5}}{\frac{49}{30}} = \frac{48}{49}$$

the difference between the correct answer and his answer

$$= \frac{48}{49} - \frac{6}{7} = \frac{6}{49}$$

Sol.174.(b) $(\sqrt{2} + \sqrt{5} - \sqrt{3}) \times k = -12$

$$\Rightarrow k = \frac{-12}{(\sqrt{2} + \sqrt{5} - \sqrt{3})}$$

$$\frac{-12}{(\sqrt{2} + \sqrt{5} - \sqrt{3})} \times \frac{(\sqrt{2} + \sqrt{5} + \sqrt{3})}{(\sqrt{2} + \sqrt{5} + \sqrt{3})}$$

$$= \frac{-12(\sqrt{2} + \sqrt{5} + \sqrt{3})}{(\sqrt{2} + \sqrt{5})^2 - (\sqrt{3})^2}$$

$$= \frac{-12(\sqrt{2} + \sqrt{5} + \sqrt{3})}{4 + 2\sqrt{10}}$$

$$= \frac{-6(\sqrt{2} + \sqrt{5} + \sqrt{3})}{2 + \sqrt{10}} \times \frac{\sqrt{10} - 2}{\sqrt{10} - 2}$$

$$= \frac{-6(\sqrt{2} + \sqrt{5} + \sqrt{3})(\sqrt{10} - 2)}{6}$$

$$= (\sqrt{2} + \sqrt{5} + \sqrt{3})(2 - \sqrt{10})$$

Sol.175.(a) $\left(1\frac{1}{3} \div 2\frac{6}{7} \text{ of } 5\frac{3}{5} \right) \div \left(6\frac{2}{5} \div \right.$

$$\left. 4\frac{1}{2} \text{ of } 5\frac{1}{3} \right) \times \left(\frac{3}{4} \times 2\frac{2}{3} \div \frac{5}{9} \text{ of } 1\frac{1}{5} \right)$$

$$= 1 + k,$$

$$= \left(\frac{4}{3} \div 16 \right) \div \left(\frac{32}{5} \div 24 \right)$$

$$\times \left(\frac{3}{4} \times \frac{8}{3} \div \frac{2}{3} \right) = 1 + k$$

$$= \left(\frac{1}{12} \right) \div \left(\frac{4}{15} \right) \times (3) = 1 + k$$

$$= \frac{5}{16} \times 3 = 1 + k$$

$$= 0.9375 = 1 + k \Rightarrow k = -0.0625$$

Clearly k lies between -0.07 and -0.06

$$\begin{aligned}\text{Sol.176.(a)} & \sqrt{10 + 2(\sqrt{6} - \sqrt{15} - \sqrt{10})} \\ &= \sqrt{10 + 2(\sqrt{2} \cdot \sqrt{3} - \sqrt{5} \cdot \sqrt{3} - \sqrt{2} \cdot \sqrt{5})} \\ &= \sqrt{(\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{5})^2 + 2(\sqrt{2} \cdot \sqrt{3} - \sqrt{5} \cdot \sqrt{3} - \sqrt{2} \cdot \sqrt{5})} \\ &= \sqrt{(\sqrt{2} + \sqrt{3} - \sqrt{5})^2} = (\sqrt{2} + \sqrt{3} - \sqrt{5})\end{aligned}$$

$$\begin{aligned}\text{Sol.177.(a)} & 0.5\overline{6} - 0.7\overline{23} + 0.3\overline{9} \times 0.\overline{7} \\ &= \frac{56-5}{90} - \frac{723-7}{990} + \frac{39-3}{90} \times \frac{7}{9} \\ &= \frac{51}{90} - \frac{716}{990} + \frac{28}{90} \Rightarrow \frac{561-716+308}{990} \\ &= \frac{153}{990} = 0.1545454 \dots \infty = 0.1\overline{54}\end{aligned}$$

Alternate method :

$$\begin{aligned}0.5\overline{6} - 0.7\overline{23} + 0.3\overline{9} \times 0.\overline{7} \\ &= 0.5\overline{6} - 0.7\overline{23} + \frac{39-3}{90} \times \frac{7}{9} \\ &= 0.5\overline{6} - 0.7\overline{23} + \frac{28}{90} \\ &= 0.5\overline{6} - 0.7\overline{23} + 0.3\overline{1} \\ &= 0.5\overline{66} - 0.7\overline{23} + 0.3\overline{11} = 0.1\overline{54}\end{aligned}$$

$$\begin{aligned}\text{Sol.178.(c)} & \sqrt{28 + 10\sqrt{3}} - \sqrt{7 - 4\sqrt{3}} \\ &= \sqrt{25 + 3 + 2 \times 5 \times \sqrt{3}} \\ &\quad - \sqrt{4 + 3 - 2 \times 2 \times \sqrt{3}} \\ &= \sqrt{(5 + \sqrt{3})^2} - \sqrt{(2 - \sqrt{3})^2} \\ &= (5 + \sqrt{3}) - (2 - \sqrt{3}) \\ &= 3 + 2\sqrt{3} = 6.46 \approx 6.5\end{aligned}$$

Sol.179.(c)

$$\begin{aligned}& \frac{2\sqrt{10}}{\sqrt{5} + \sqrt{2} - \sqrt{7}} - \sqrt{\frac{\sqrt{5}-2}{\sqrt{5}+2}} - \frac{3}{\sqrt{7}-2} \\ &= \frac{2\sqrt{10}}{\sqrt{5} + \sqrt{2} - \sqrt{7}} \times \frac{\sqrt{5} + \sqrt{2} + \sqrt{7}}{\sqrt{5} + \sqrt{2} + \sqrt{7}} - \\ &\quad \sqrt{\frac{\sqrt{5}-2}{\sqrt{5}+2}} \times \frac{\sqrt{5}-2}{\sqrt{5}-2} - \frac{3}{\sqrt{7}-2} \\ &\quad \times \frac{\sqrt{7}+2}{\sqrt{7}+2} \\ &= \frac{(2\sqrt{10})(\sqrt{5} + \sqrt{2} + \sqrt{7})}{(\sqrt{5} + \sqrt{2})^2 - (\sqrt{7})^2} \\ &\quad - \sqrt{\frac{(\sqrt{5}-2)^2}{(\sqrt{5})^2 - 2^2}} - \frac{3(\sqrt{7}+2)}{(\sqrt{7})^2 - 2^2} \\ &= \frac{(2\sqrt{10})(\sqrt{5} + \sqrt{2} + \sqrt{7})}{7 + 2\sqrt{10} - 7} \\ &\quad - (\sqrt{5} - 2) - (\sqrt{7} + 2) \\ &= \sqrt{5} + \sqrt{2} + \sqrt{7} - \sqrt{5} + 2 - \sqrt{7} - 2 = \sqrt{2}\end{aligned}$$

Sol.180.(a)

$$\begin{aligned}& 9\frac{4}{9} \div 11\frac{1}{3} \text{ of } \frac{1}{6} + (1\frac{1}{3} \times 1\frac{4}{5} \div \frac{3}{5}) \\ & \times 2\frac{1}{6} \text{ of } \frac{2}{3} \div \frac{4}{3} \text{ of } \frac{2}{3} \\ &= \frac{85}{9} \div \frac{34}{18} + (\frac{4}{3} \times \frac{9}{5} \div \frac{3}{5}) \times \frac{13}{9} \div \frac{8}{9} \\ &= 5 + 4 \times \frac{13}{8} = \frac{23}{2}\end{aligned}$$

$$\text{Required difference} = \frac{77}{4} - \frac{23}{2} = 7\frac{3}{4}$$

$$\text{Sol.181.(d)} 99\frac{95}{99} \times 99 - 95$$

$$\begin{aligned}& \Rightarrow 99 \times 99 + \frac{95}{99} \times 99 - 95 \\ & \Rightarrow 9801 + 95 - 95 = 9801\end{aligned}$$

$$\text{Sol.182.(a)} 5 \div 10 \text{ of } 10 \times 4 + 4 \div 4 \text{ of } 4$$

$$\begin{aligned}& \times 10 - (10 - 4) \div 16 \times 4 \\ &= 5 \div 100 \times 4 + 4 \div 16 \times 10 - \frac{3}{8} \times 4 \\ &= \frac{1}{20} \times 4 + \frac{5}{2} - \frac{3}{2} \Rightarrow 0.2 + 1 = 1.2\end{aligned}$$

$$\begin{aligned}\text{Sol.183.(a)} & \frac{(3\frac{1}{5} - \frac{3}{5}) \div \frac{8}{5}}{1\frac{1}{7} \div \{\frac{6}{7} - (\frac{1}{7} \div \frac{1}{5})\}} \\ &= \frac{(\frac{13}{5}) \div \frac{8}{5}}{\frac{1}{7} \div (\frac{6}{7} - \frac{5}{7})} \Rightarrow \frac{\frac{13}{8}}{\frac{1}{7} \div \{\frac{6}{7} - \frac{5}{7}\}} \\ &= \frac{13}{8 \times 8} = \frac{13}{64}\end{aligned}$$

$$\text{Sol.184.(c)} \frac{8}{9} \text{ of } (5\frac{1}{4} \div 2\frac{1}{3} \text{ of } 4) \div$$

$$\begin{aligned}& (8 \div \frac{2}{3} \text{ of } \frac{4}{5}) \text{ of } (8 \times \frac{2}{3} \div \frac{4}{5}) \\ &= \frac{8}{9} \text{ of } (\frac{21}{4} \div \frac{28}{3}) \div (8 \div \frac{8}{15}) \text{ of } (8 \times \frac{5}{6}) \\ &= \frac{8}{9} \text{ of } \frac{9}{16} \div 15 \text{ of } \frac{20}{3} \\ &= \frac{1}{2} \div 100 = \frac{1}{200}\end{aligned}$$